Final Version

Executive Summary

1. Introduction and Strategic Context

1.1 Background: Batteries as a Strategic Industrial Sector

Batteries matter because they enable electrification where direct, real-time grid supply is either not available, not stable enough, or not flexible enough. In mobility, they are the core enabling technology for electric cars, buses, trucks, two-wheelers, and three-wheelers. In power systems, they are becoming the balancing asset that helps solar and wind move from intermittent generation to dispatchable electricity. In data centres and other digital infrastructure, storage and UPS systems are increasingly important because power quality, continuity, and flexibility are becoming strategic issues as electricity demand rises. Electricity demand from data centres is projecte d to more than double to around 945 TWh by 2030, while global battery storage deployment reached 108 GW in 2025.IEA

The scale-up is already visible. Electric vehicles remained the primary driver of global battery deployment in 2025, accounting for over 70% of the total, with EV battery deployment reaching 1.2 TWh IEA. Battery prices also continued to decline in 2025, although mineral-market volatility remains a live risk. In parallel, stationary battery storage is scaling rapidly: the IEA describes battery storage as the fastest-growing power technology today, with lithium iron phosphate batteries accounting for around 90% of deployments in 2025. This combination mobility plus power has made batteries a foundational industrial platform rather than a single end product.

Batteries are also strategic because they are inseparable from critical minerals and industrial security. Refined critical mineral production remains highly concentrated in a small number of countries IEA. Now, roughly 70–75% of projected supply growth for refined lithium, nickel, cobalt, and rare earth elements comes from today's top three producers, and almost 95% of projected growth in battery-grade spherical and synthetic graphite comes from China.

1.2 Objective and Scope of the Report

The objective of this report is to provide a client-ready, layman-friendly, and evidence-based understanding of the battery ecosystem for four core audiences: policy makers designing industrial and circular-economy interventions; industry participants across manufacturing, recycling, logistics, and energy; investors assessing value pools and risks; and MSMEs seeking realistic entry points in a growing ecosystem. The report is designed to explain the "what," "why," and "so what" of batteries in simple language without losing strategic depth.

The scope covers **rechargeable batteries only**, with a focus on the full life cycle: raw materials, refining, active materials, cell manufacturing, module and pack assembly, first-life applications,

second-life pathways, recycling, and material recovery. A World vs India framing is used throughout to compare global leadership patterns with India's current gaps and opportunities. The report includes battery chemistries, applications, recycling routes, policy context, and future technology directions. It excludes deep electrochemistry, detailed plant engineering, valuation of individual companies, and primary non-rechargeable batteries except where needed for context.

1.3 Key Questions Addressed

This report seeks to answer the following strategic questions:

- **What makes batteries a strategic industrial sector?**
  Why have batteries become central to electric mobility, renewable energy integration, industrial electrification, energy security and economic competitiveness?
- **How has the global battery ecosystem evolved?**
  What technological, economic and policy developments have transformed batteries from a niche product into a global manufacturing industry?
- **How does the battery value chain operate?**
  What are the key stages of the battery ecosystem—from critical mineral extraction and refining through cell manufacturing, battery-pack assembly, deployment, second-life applications and recycling—and where is value created along the chain?
- **What are the major battery chemistries and technologies?**
  How do lead-acid, lithium-ion and emerging battery technologies differ in terms of performance, safety, cost, applications and critical mineral requirements?
- **What are the major trends shaping the global battery market?**
  How are electric vehicles, battery energy storage systems, industrial electrification, manufacturing concentration and supply-chain risks influencing the future of the battery industry?
- **Why is battery recycling becoming strategically important?**
  What role do collection systems, refurbishment, second-life applications, black mass recovery and circular economy principles play in strengthening resource security and reducing dependence on primary raw materials?
- **What is India's current position within the global battery ecosystem?**
  What are India's strengths, constraints and emerging opportunities across battery demand, manufacturing, policy, critical minerals and recycling?
- **How prepared is India's manufacturing ecosystem for future battery demand?**
  What gaps exist across domestic manufacturing, supply chains, technology, infrastructure, skills and industrial policy, and what measures are required to strengthen India's competitiveness?

- **What lessons can India draw from global experience?**
  How can international manufacturing models, industrial ecosystems and circular economy approaches inform India's long-term battery strategy?

Collectively, these questions aim to provide policymakers, industry stakeholders, investors and researchers with a comprehensive understanding of the battery ecosystem and identify the strategic priorities required to strengthen India's position within the global battery value chain.

1.4 Report Structure

The report is organised into nine thematic chapters that progressively examine the battery ecosystem from its technological evolution to India's manufacturing opportunities and circular economy potential.

Chapter 1 introduces the strategic importance of batteries, defines the objectives and scope of the report, and outlines the key questions addressed.

Chapter 2 traces the evolution of rechargeable battery technologies, explaining how advances in battery chemistry, manufacturing and policy transformed batteries from a niche technology into a strategic industrial sector supporting electric mobility, renewable energy integration and industrial electrification.

Chapter 3 examines the global battery value chain, covering critical mineral extraction, refining, active material production, cell manufacturing, module and pack assembly, downstream applications and end-of-life management. The chapter explains where economic value is created and highlights the strategic importance of each stage within the battery ecosystem.

Chapter 4 analyses the major battery chemistries currently used in commercial applications, including lead-acid, lithium-ion variants and emerging technologies such as sodium-ion and solid-state batteries. It compares these technologies in terms of performance, cost, safety, energy density, mineral dependency and suitability for different end-use applications.

Chapter 5 provides an overview of the global battery market, examining demand projections, manufacturing capacity, regional competitiveness, leading industry participants, evolving business models, cost trends and supply-chain risks. The chapter also explores how different countries have developed globally competitive battery manufacturing ecosystems.

Chapter 6 focuses on battery recycling and circular economy pathways. It discusses battery life-cycle management, collection systems, testing and grading, second-life applications, recycling technologies, black mass recovery, economics, regulatory frameworks and the role of circularity in strengthening resource security and reducing dependence on primary raw materials.

Chapter 7 assesses India's battery ecosystem by analysing domestic demand growth, manufacturing capabilities, policy initiatives, critical mineral availability, recycling readiness and existing industrial gaps. The chapter evaluates India's preparedness to develop a competitive

battery manufacturing ecosystem and identifies the key constraints affecting domestic value addition.

Chapter 8 examines the policy and industrial ecosystem required to accelerate battery manufacturing in India. It evaluates national and state-level initiatives, manufacturing incentives, industrial infrastructure, localisation strategies, research and development priorities, skill development and investment frameworks necessary for building an integrated battery value chain. The chapter also analyses the role of coordinated industrial policy in strengthening India's competitiveness across upstream, midstream and downstream manufacturing.

Chapter 9 explores future opportunities and strategic pathways for India's battery industry. It synthesises global best practices, emerging technology trends, circular economy approaches and manufacturing strategies to identify priority areas for strengthening domestic capabilities, enhancing supply-chain resilience and increasing India's participation in global battery value chains.

The report concludes by synthesising the major findings and presenting a strategic roadmap for developing an integrated battery manufacturing ecosystem in India. Rather than viewing batteries solely through the lens of electric vehicles, the conclusion emphasises the need to strengthen capabilities across the entire value chain—including critical minerals, advanced manufacturing, innovation, recycling and regional industrial clusters—to maximise domestic value addition, improve supply-chain resilience and position India as a leading global battery manufacturing hub

2. Evolution of the Global Battery Ecosystem

2.1 Early Battery Systems: Lead-Acid and Nickel-Based Technologies

The modern rechargeable battery economy began with lead-acid. The Nobel scientific background for the 2019 Chemistry Prize notes that Wilhelm Sinsteden studied the chemistry in 1854 and Gaston Planté demonstrated the lead-acid battery in 1859–1860. Its enduring strength was straightforward: low cost, high surge current, and simple manufacturing. Its main limitation was also clear from the beginning: low energy density, which made it heavy and less suitable for applications where weight and size matter. That trade-off is why lead-acid still survives in starter batteries, UPS systems, and backup power, even after more advanced chemistries emerged.

Nickel-based batteries were the next major step. In 1899, Waldemar Jungner described nickel-iron and nickel-cadmium batteries; these alkaline systems later became predecessors to nickel-metal hydride, which the Nobel background identifies as commercialized in 1989. Compared with lead-acid, nickel systems were more robust and better suited to portable and repetitive-use applications, but they were generally costlier and less attractive for very large-scale transport or stationary use once lithium-ion matured.

Chemistry	Cost	Energy density	Cycle life	Recycling status	Uses
Lead-acid	Low	Low	Low to medium	Highly established	Starter batteries, telecom backup, UPS
Nickel-based	Medium	Medium	Medium	Established, but cadmium raises toxicity concerns in Ni-Cd	Power tools, emergency systems, early hybrids
Lithium-ion	Higher upfront cost	High	High	Growing but still maturing at scale	Consumer electronics, EVs, BESS

Indicative comparison that widely used technical battery characteristics references.

2.2 Rise of Lithium-Ion Batteries

Lithium-ion emerged because earlier lithium-metal designs solved energy density but not safety. The Nobel background records that Whittingham's early rechargeable lithium system used a lithium-metal anode and Titanium Disulfide (TiS_2) cathode; it demonstrated reversibility, 2.5 V cell potential, and even cycling over 1,100 times in laboratory conditions. But lithium dendrites created short-circuit and fire risks, which halted commercial progress.

The next breakthrough came from Goodenough, who showed in 1979–1980 that Lithium cobalt oxide (LiCoO2) could work as a high-voltage cathode. The Nobel background reports a potential of roughly 4–5 V versus Li+/Li, which was a major jump over earlier systems. Yoshino then solved the anode side in 1985 by identifying carbonaceous petroleum-coke-based materials that could repeatedly host lithium ions more safely than lithium metal. That combination led to the first commercial lithium-ion battery in 1991.

Today, the center of gravity has moved from invention to scale. The global battery industry is dominated by companies headquartered in China, Korea, and Japan, with Chinese producers supplying nearly three-quarters of EV batteries deployed in 2025 IEA. In simple terms: innovation created the market, but manufacturing ecosystems now decide who captures value.

2.3 EVs and Energy Storage as Battery Demand Accelerators

The battery industry became strategically important only when two demand waves arrived at the same time: electric vehicles and stationary storage. EV battery deployment reached 1.2 TWh in 2025, up almost 30% from 2024; electric vehicles still accounted for more than 70% of total battery deployment, and electric trucks were the fastest-growing segment IEA. At the same time, the

power sector added 108 GW of battery-storage capacity in 2025, with around 80% of those additions at utility scale.

This matters because it changes the economics of the entire sector. EVs pull scale, storage pulls duration and cycling, and both together expand the market for LFP, NMC, graphite, separators, electrolytes, software, power electronics, and recycling. In India, the same direction is visible in policy even if domestic scale remains smaller: the ACC PLI scheme is trying to seed domestic manufacturing, while Battery Waste Management Rules, 2022 already assume that collection, refurbishment, and recycling will become integral parts of the market.

2.4 Strategic Shift: From Battery Product to Battery Ecosystem

The most important strategic shift is that batteries can no longer be analyzed as a single finished product. The industry has expanded into a full value chain: mining, refining, cathode and anode materials, cells, modules, packs, thermal management, BMS, system integration, use-phase analytics, second-life use, and recycling. Structural concentration remains high across the battery value chain, particularly in LFP materials and graphite anodes, with heavy dependence on Chinese manufacturing capacity and expertise. IEA

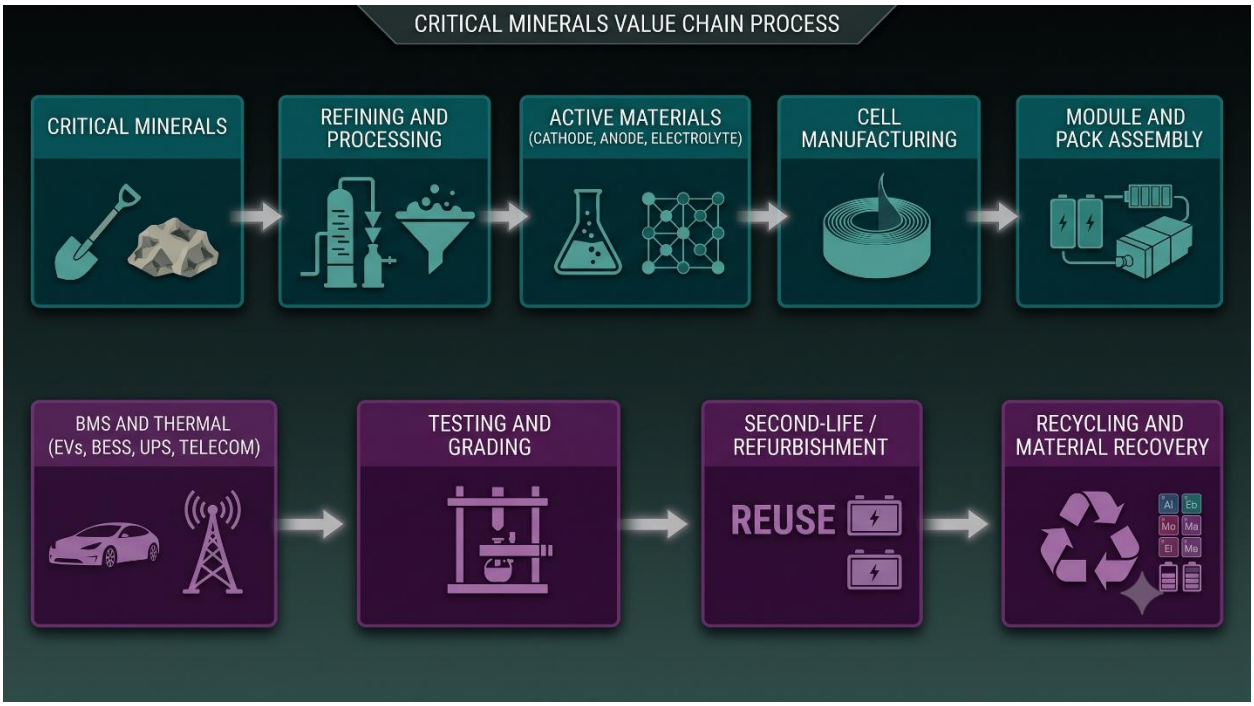


Figure source: synthesized from IEA battery value-chain and recycling analysis; India's rules explicitly cover collection, refurbishment, recycling, and second life.

The policy implication is straightforward. If governments focus only on final cell assembly, they capture only a fragment of value. If they develop upstream materials, downstream system integration, and end-of-life recovery together, they build a true battery ecosystem. For India, this is especially relevant because Battery Waste Management Rules already define refurbishment and

repurposing, and the ACC PLI scheme pushes domestic value addition. That makes the battery sector not only a clean-energy story, but also an industrial-development story.

Year	Milestone	Strategic significance
1859–1860	Planté demonstrates lead-acid	First practical rechargeable battery; foundation for automotive and backup uses
1899	Jungner describes Ni-Cd and Ni-Fe	Better durability and portability; precursor to later NiMH
1976	Whittingham demonstrates lithium intercalation battery	First modern proof that lithium-based rechargeable systems could work
1979–1980	Goodenough team identifies LiCoO2 cathode	High-voltage cathode made higher-energy batteries possible
1985	Yoshino develops carbon-anode lithium-ion cell	Safety and manufacturability breakthrough
1991	Commercial lithium-ion battery released	Battery industry moves from lab science to mass market
2020	EV plus storage demand surges	Batteries become a strategic industrial ecosystem

Summary table of evolution of rechargeable battery

3. Global Battery Value Chain Structure

3.1 Overview of the Battery Value Chain

A battery moves through seven practical stages. First come mining and raw-material extraction. Second comes refining and chemical conversion into battery-grade inputs such as lithium carbonate, lithium hydroxide, nickel sulphate, cobalt sulphate and purified graphite. Third comes precursor and active material production for cathodes and anodes, together with separators and electrolytes. Fourth comes cell manufacturing. Fifth comes module, pack and battery management system integration. Sixth comes deployment into end-use markets such as EVs, stationary storage, telecom, UPS, consumer devices and industrial equipment. Seventh comes end-of-life management: collection, diagnostics, second life, recycling and safe disposal. Batteries are part of a broader ecosystem encompassing critical minerals, manufacturing, use, and recycling, rather than a stand-alone industry. IEA

3.2 Upstream: Critical Minerals and Raw Materials

The upstream layer is defined by a familiar set of minerals, but each plays a different role. Lithium, nickel, cobalt, manganese, graphite and phosphate are central to lithium-ion chemistries; copper is

critical for current collectors, cabling and electrical systems; lead remains essential for the large installed base of lead-acid batteries in automotive and backup applications. The supply-risk issue is not just whether reserves exist; it is whether mine output and, more importantly, processing are concentrated in a small number of countries.

https://pubs.usgs.gov/periodicals/mcs2026/

Mineral	2025 estimated world mine production	World reserves	Top mine producer in 2025	Strategic concentration note
Lithium	~290 kt Li	37 Mt Li	Australia — 92 kt, ~32%	Mining diversified; refining concentrated in China
Cobalt	~310 kt Co	12 Mt Co	DRC — 230 kt, ~74%	DRC dominates mining: China leads refining
Nickel	~3.9 Mt Ni	>140 Mt Ni	Indonesia — 2.6 Mt, ~67%	Indonesia dominates mining and refining
Manganese	~20 Mt Mn content	1.8 Bt Mn content	South Africa — 7.6 Mt, ~38%	Battery-grade processing remains concentrated
Natural graphite	~1.8 Mt	310 Mt	China — 1.4 Mt, ~78%	China dominates mining and processing
Copper	~23 Mt Cu	980 Mt Cu	Chile — 5.3 Mt, ~23%	Mining diversified; refining concentrated in China
Phosphate rock	~250 Mt	73 Bt	China — 110 Mt, ~44%	China leads production; Morocco holds most reserves
Lead	~4.5 Mt Pb	95 Mt Pb	China — 1.9 Mt, ~42%	Mature market with strong recycling systems

The key structural insight is that the geographical distribution of mineral reserves is generally broader than the concentration of processing capacity. Lithium reserves are spread across Australia, Chile, Argentina and several emerging producers; nickel reserves exist across multiple regions; and phosphate rock is abundant, with Morocco holding the largest share of global reserves. However, downstream processing remains much more concentrated. China dominates lithium refining, cathode materials, natural graphite processing and a significant share of copper refining, while Indonesia has become the leading centre for nickel mining and processing. Cobalt follows a similar pattern, with mining concentrated in the Democratic Republic of the Congo and refining led by China. This explains why battery industrial policy is increasingly focused not only on

securing mineral resources, but also on building domestic refining, purification and intermediate-material capabilities.

3.3 Midstream: Refining, Processing and Cell Components

The midstream is where ores and concentrates become battery-grade inputs, and it is the least visible but most strategically powerful part of the chain. Lithium must be converted into battery-grade carbonate. Nickel and cobalt must be refined into sulphates. Graphite must be purified and shaped into spherical graphite for anodes. Cathode production then combines metal salts into precursor cathode active materials and finally cathode active materials such as NMC or LFP. Separators and electrolytes sit alongside these materials as essential enabling inputs. This step is capital-intensive, chemistry-intensive and qualification-heavy; it is also where quality failures can cascade downstream into cell performance and safety issues.

3.4 Cell Manufacturing, Module and Pack Assembly

Cell manufacturing has scaled with extraordinary speed. Global lithium-ion battery manufacturing capacity exceeded 4 TWh by the end of 2025, representing a year-on-year increase of roughly 30% IEA. China still accounted for more than 80% of global nameplate capacity, while the European Union and the United States each accounted for roughly 6-7%. China, Europe and North America together represented about 95% of output between 2023 and 2025.

This is the stage most visible to policymakers because gigafactories are large, investible and politically salient. Yet even here, producer nationality matters as much as plant location. IEA The Chinese, Korean, and Japanese firms continue to supply nearly all battery cells globally, with Chinese producers accounting for almost 75% of global electric vehicle battery deployment in 2025. Pack assembly is more geographically distributed because it must be close to the vehicle or storage application, but packs still depend on cell and component ecosystems. Battery management systems, cooling design, pack structure and software integration are therefore the practical bridge between upstream chemistry and downstream application value.

Manufacturer scale indicator	**Headquarters**	**Comparable scale metric**	**Chemistry / positioning**
CATL	China	464.7 GWh EV batteries installed in 2025	Broad portfolio; strong in LFP, NMC and storage
BYD / FinDreams	China	194.8 GWh in 2025	LFP-heavy, integrated with vehicle manufacturing

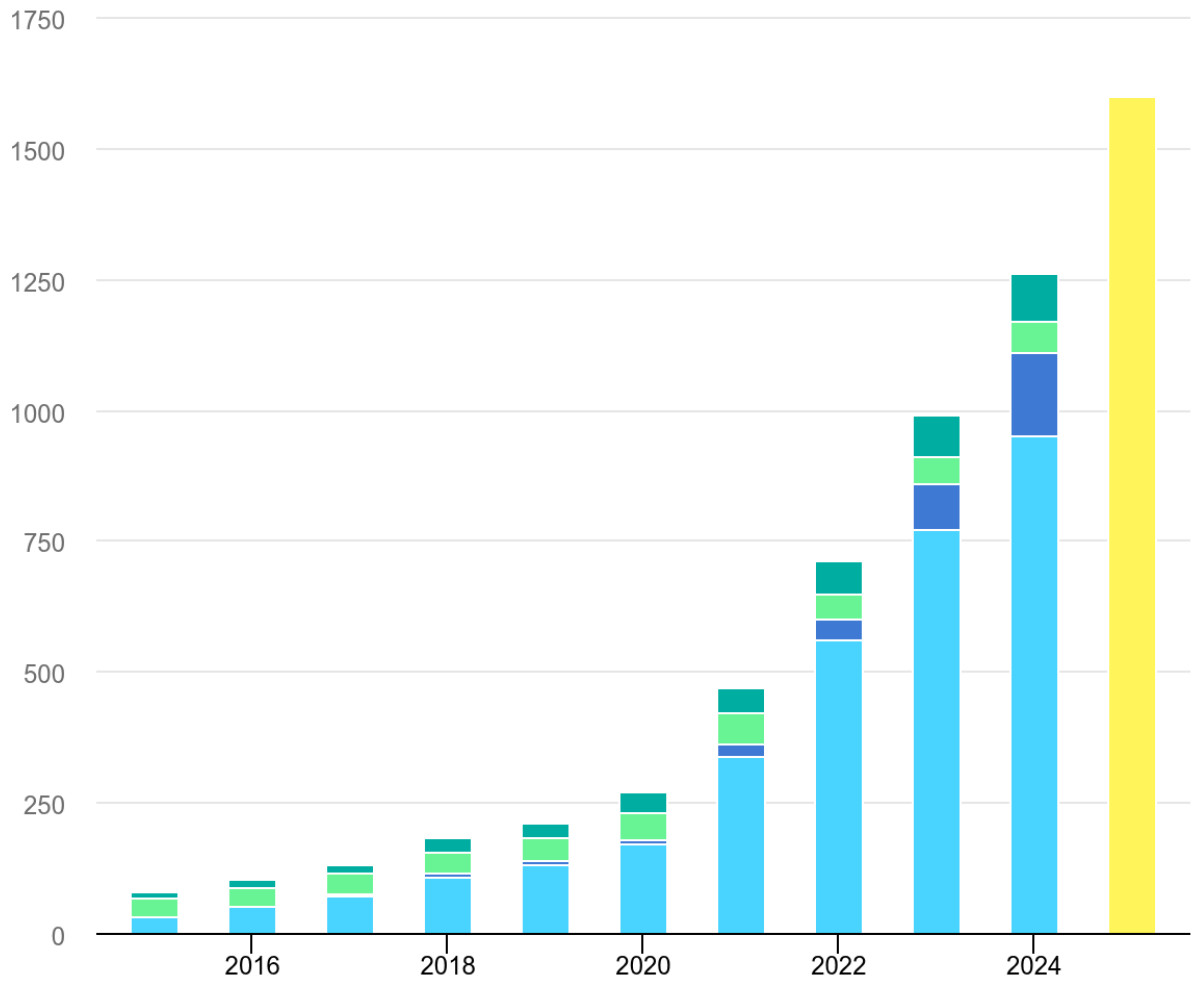
LG Energy Solution	Korea	108.8 GWh in 2025	Historically NMC/NCx-heavy; expanding LFP

Table: Leading Global Battery Manufacturers and Their Market Positioning

Chemistry specialization also matters. IEA LFP is the prevailing chemistry in China and in stationary storage, while NMC remains more common in Europe and North America.

3.5 Downstream Demand: EVs, Storage and Industrial Applications

Demand is now anchored by two uses: electric mobility and stationary storage. IEA EVs accounted for more than 70% of total battery deployment in 2025, with EV battery deployment reaching 1.2 TWh. It also notes that EVs and stationary storage together account for around 90% of today's lithium-ion battery market. Storage is rapidly becoming the second anchor market: 108 GW of new battery storage capacity was deployed globally in 2025, up 40% from 2024, and LFP represented around 90% of battery storage deployments.



IEA (2026), Lithium-ion battery deployment by application, 2015-2025, IEA, Paris https://www.iea.org/data-and-statistics/charts/lithium-ion-battery-deployment-by-application-2015-2025

Approximate 2025 lithium-ion demand mix

EVs		>70%
Stationary storage		~20%
Other applications		<10%

Regional demand drivers are different. China is the dominant market in both EVs and storage and was already 55% of annual global battery-storage additions in 2023. The United States is seeing especially rapid stationary storage growth, with battery storage accounting for one-third of battery deployment in 2025, aided by grid needs and data-centre growth. Europe's demand is shaped by EV policy, industrial sourcing rules and higher behind-the-meter storage adoption, while emerging markets including India, Southeast Asia and Latin America are still small in battery deployment terms but are expected to gain share quickly through 2035.

3.6 End-of-Life Segment: Collection, Reuse, Recycling and Disposal

End-of-life is becoming a strategic segment, but it is not yet a large substitute for mining. The IEA notes that recycling will take about a decade to materially reduce primary mineral demand because feedstock volumes are still limited. Reuse and repurposing are technically feasible, especially for less demanding stationary applications, but economics are difficult because dismantling, safety assurance and competition from new batteries all matter. In India, the Battery Waste Management Rules, 2022 create an EPR system under which producers must ensure collection and recycling and/or refurbishment; collected waste batteries are not to be sent for landfilling or incineration.

Pathway	What it means	Best-fit use case	Main benefit	Main limitation
Reuse / second life	Battery is redeployed in a less demanding application	EV batteries moving into stationary storage	Extends useful life and delays recycling	Testing, repackaging and safety certification are costly
Refurbishment	Repair or replacement of defective modules/cells to	Industrial and EV packs with recoverable value	Preserves asset value and supports service markets	Requires diagnostics and traceability

	restore serviceability			
Recycling	Recovery of metals/materials through pyro, hydro or emerging direct routes	End-of-life batteries with low reuse value	Returns lithium, nickel, cobalt, graphite and other materials to the chain	Short-term feedstock still limited; economics vary by chemistry
Disposal	Final treatment without material recovery	Only residual hazardous fractions	Necessary for unrecoverable waste	Destroys value and creates environmental burden

4. Battery Chemistries and Technology Landscape

4.1 Overview of Battery Chemistry Selection

Chemistry choice is never just a technical decision. It determines how much a battery costs, how safe it is, how far a vehicle can travel, how long a storage asset can cycle, and which minerals a country becomes dependent on. The industry therefore does not converge on one “best” chemistry; it optimises different chemistries for different use cases. That is why LFP, NMC, NCA, LMO, LTO and emerging sodium-ion or solid-state systems coexist rather than fully replace one another.

Chemistry selection is a multi-criteria optimisation problem, not a single "best" answer. The parameters that matter, and why each matters industrially:

Criterion	Why it matters	Which chemistries win on it
Gravimetric energy density (Wh/kg)	Determines vehicle range and payload for a given battery weight	NCA, NMC
Volumetric energy density (Wh/l)	Determines how much space the battery pack occupies	NCA, NMC
Power density (W/kg)	Determines acceleration and fast-charge capability	LTO, NMC
Cycle life (number of full charge-discharge cycles before capacity fades below ~80%)	Determines usable lifetime, warranty cost, and second-life potential	LFP, LTO
Calendar life (ageing independent of use)	Matters for stationary storage and low-utilization fleets	LFP
Thermal stability / safety	Determines fire risk and the extent of thermal-management engineering needed	LFP
Low-temperature performance	Determines usability in cold climates without heating overhead	Sodium-Ion, LFP (weaker)
Cost per kWh	Determines affordability and mass-market viability	LFP, Sodium-Ion (at scale)
Critical-mineral dependency	Determines supply-chain exposure and geopolitical risk	LFP (No Ni/Co), Sodium-Ion (No Li Heavy Dependency

4.2 Lead-Acid Batteries

Lead-acid remains the legacy chemistry with enduring relevance. Its strengths are low upfront cost, mature manufacturing, high cranking power and established recycling systems. Its weaknesses are equally clear: very low energy density and lower deep-cycle performance compared with lithium-ion. As a result, lead-acid remains strong in starter batteries, backup power and low-cost storage, but it is poorly suited to modern mainstream EV traction.

4.3 Lithium-Ion Chemistries: LFP, NMC, NCA, LMO and LTO

LFP has become the scale chemistry because it lowers cost, avoids nickel and cobalt, improves thermal stability and delivers strong cycle life, even if its energy density is lower than nickel-rich chemistries. IEA LFP represented nearly half of the global EV battery market in 2024; in India, as in Southeast Asia and Brazil, LFP exceeded 50% of electric-car battery sales. LFP battery packs were >40% cheaper per kWh than NMC in 2025, and LFP now accounts for over 55% of global EV battery deployment (up from ~50% in 2024). In emerging markets outside China, LFP already

powers two-thirds of EV sales — double its 2023 share. High-nickel NMC (721/811) and NCA together made up ~80% of 2025 EV battery deployment among cobalt-containing chemistries — but use comparatively little cobalt per cell versus older NMC111-era batteries. LMO occupies a more limited role, often in blended designs, while LTO remains a niche chemistry suited to very fast charging and exceptional cycle life rather than maximum range.

4.4 Emerging Technologies: Sodium-Ion, Solid-State and Flow Batteries

IEA sodium-ion, solid-state, lithium-sulphur, iron-air and redox-flow batteries as the main frontier technologies under development. Sodium-ion aims to reduce lithium dependence and is moving into early commercialisation: CATL launched its Naxtra sodium-ion brand in 2025, with mass production planned from December 2025 and reported energy density of 175 Wh/kg. Solid-state batteries offer the promise of higher range and safety, but the IEA still places them at large pilot stage and expects initial commercial volumes to remain limited even if first mass production begins in 2027-2028. Flow batteries are different again: they are less about mobility and more about long-duration stationary storage.

Chemistry	**Relative cost**	**Relative safety**	**Relative energy density**	**Mineral dependency profile**	**Best-fit applications**
Lead-acid	Low	Medium	Low	High dependence on lead; mature recycling	Starter batteries, telecom backup, low-cost stationary uses
LFP	Low	High	Medium	Lithium and graphite, but no nickel/cobalt	Mass-market EVs, buses, stationary storage
NMC	Medium to high	Medium	High	Lithium, graphite, nickel and cobalt exposure	Passenger EVs where range matters
NCA	Medium to high	Medium to lower than LFP	Very high	High nickel dependence, some cobalt	Premium long-range EVs
LMO	Medium	Medium	Medium to low	Lithium, manganese, graphite	Power-oriented uses, often blended with other chemistries

Chemistry	Relative cost	Relative safety	Relative energy density	Mineral dependency profile	Best-fit applications
LTO	High	High	Very low	Lithium and titanium	Fast-charging fleets, high-cycle industrial uses
Sodium-ion	Potentially low at scale	High	Medium to low	Reduced lithium dependence; more abundant feedstocks	Stationary storage, cost-sensitive mobility, cold-climate use cases
Solid-state	High today, commercial cost still uncertain	Potentially high	Potentially very high	Varies by design; often still lithium-based	Future high-range mobility, aviation/marine niches

Source note: qualitative matrix synthesised from IEA technology assessment and commercial chemistry characteristics

4.5 Chemistry Comparison: Cost, Safety, Energy Density and Mineral Dependency

A practical rule emerges from the evidence. If the priority is low cost, long life and operational safety, LFP tends to win. If the priority is compactness and range, NMC and NCA still hold an advantage. If the priority is rapid charge and cycle endurance, LTO can justify its niche despite low energy density. If the priority is reducing lithium exposure, sodium-ion becomes attractive—though IEA notes that it still needs either higher energy density or more favourable market conditions to compete directly with LFP on price per kWh.

4.6 Strategic Takeaway: Chemistry Choices as Industrial Strategy Choices

Chemistry determines more than performance; it determines supply-chain dependence, recycling economics, localisation feasibility and trade risk. An LFP-led strategy, for example, reduces nickel and cobalt exposure but increases dependence on Chinese LFP-related component ecosystems today. A nickel-rich strategy raises energy density but deepens exposure to cobalt and nickel supply concentration. A sodium-ion option can lower lithium dependence, but only once manufacturing scale and performance improve further.

5. Global Battery Market Landscape

5.1 Global Battery Demand Outlook

Global battery demand is increasing rapidly, driven primarily by the growth of electric vehicles (EVs) and stationary energy storage. In 2024, battery demand in the energy sector surpassed 1 TWh, approximately three times the level recorded in 2021[1]. Future projections consistently indicate strong growth. The International Energy Agency (IEA) projects battery demand to exceed 3 TWh by 2030 under its Stated Policies Scenario (STEPS), while McKinsey Battery Insights forecasts total lithium-ion battery demand to reach 4.2 TWh by 2030 and 6.8 TWh by 2035[2]. Similarly, the International Renewable Energy Agency (IRENA) projects EV battery demand to exceed 4.3 TWh by 2030 under its 1.5°C scenario[3]. Collectively, these projections indicate that global battery demand is expected to increase several-fold over the coming decade, even under the most conservative scenarios. he prior year's IEA edition projected 1,600 GWh for 2025 — actual deployment came in at 1,200 GWh, about 25% lower[4].

Source (scenario)	2025	2030	2035
IEA (Global EV Outlook 2025, STEPS)	1,600	>3,000	–
McKinsey "Battery 2035" (2026)	1,600	4,200	6,800
IRENA (1.5°C scenario, EV only)	–	4,300	–
IEA GEO 2026 (CPS/STEPS)	1200	~3000	4,000 (CPS) – 4,900 (STEPS)

5.2 Key Demand Drivers: EVs, BESS and Industrial Electrification

Global battery demand is increasingly being shaped by three interconnected markets: electric vehicles (EVs), battery energy storage systems (BESS) and the broader electrification of industrial activities. EVs remain the dominant source of demand, accounting for more than 70% of global battery deployment in 2025, when EV battery deployment reached approximately 1.2 TWh almost 30% higher than in 2024[4]. Demand is also expanding beyond passenger cars, with battery deployment in electric trucks more than doubling during 2025 as freight transport begins to electrify. Electric trucks are the fastest-growing demand segment within this total — truck battery demand more than doubled in 2025 alone (largely China-driven) and is projected to rise from 8% of total EV battery deployment today to around 10% by 2030-2035[5]. In parallel, BESS is emerging

[1] https://www.iea.org/reports/global-ev-outlook-2025/electric-vehicle-batteries
[2] https://www.mckinsey.com/features/mckinsey-center-for-future-mobility/our-insights/battery-2035-building-new-advantages
[3] https://www.irena.org/Publications/2024/Sep/Critical-materials-Batteries-for-electric-vehicles
[4] https://www.iea.org/reports/global-ev-outlook-2026
https://www.iea.org/reports/global-ev-outlook-2026/electric-vehicle-batteries
[5] https://www.iea.org/reports/global-ev-outlook-2026/electric-vehicle-batteries

as the fastest-growing power-sector application for batteries, driven by the need to integrate variable renewable energy, manage peak demand, reduce grid congestion and improve electricity reliability. Global battery-storage additions reached 108 GW in 2025, representing a 40% year-on-year increase[6]. Industrial electrification is creating an additional demand layer as businesses and manufacturing facilities adopt behind-the-meter storage[7], renewable-energy systems and electrically powered transport and equipment to reduce fuel consumption, manage electricity costs and strengthen energy resilience. Together, these applications are transforming batteries from a primarily automotive component into critical infrastructure supporting mobility, electricity systems and industrial decarbonisation. International energy agency battery.[8]

5.3 Global Manufacturing Capacity and Regional Concentration

Installed capacity: World cell-manufacturing capacity has risen sharply. In 2025 it reached ~3.9 TWh/year (3909 GWh) roughly triple global battery demand and is expanding.

Headquarters	EV Battery Capacity (GWh/yr)	BESS Capacity (GWh/yr)	Total Capacity (GWh/yr)
China	3,170	560	3,730
European Union	269	10	279
North America	291	40	331
Japan	49	2	51
Korea	73	3	76
Rest of World	57	21	78

Battery cell manufacturing capacity by region, 2025 (approx.) and notable projects. (Data from IEA)

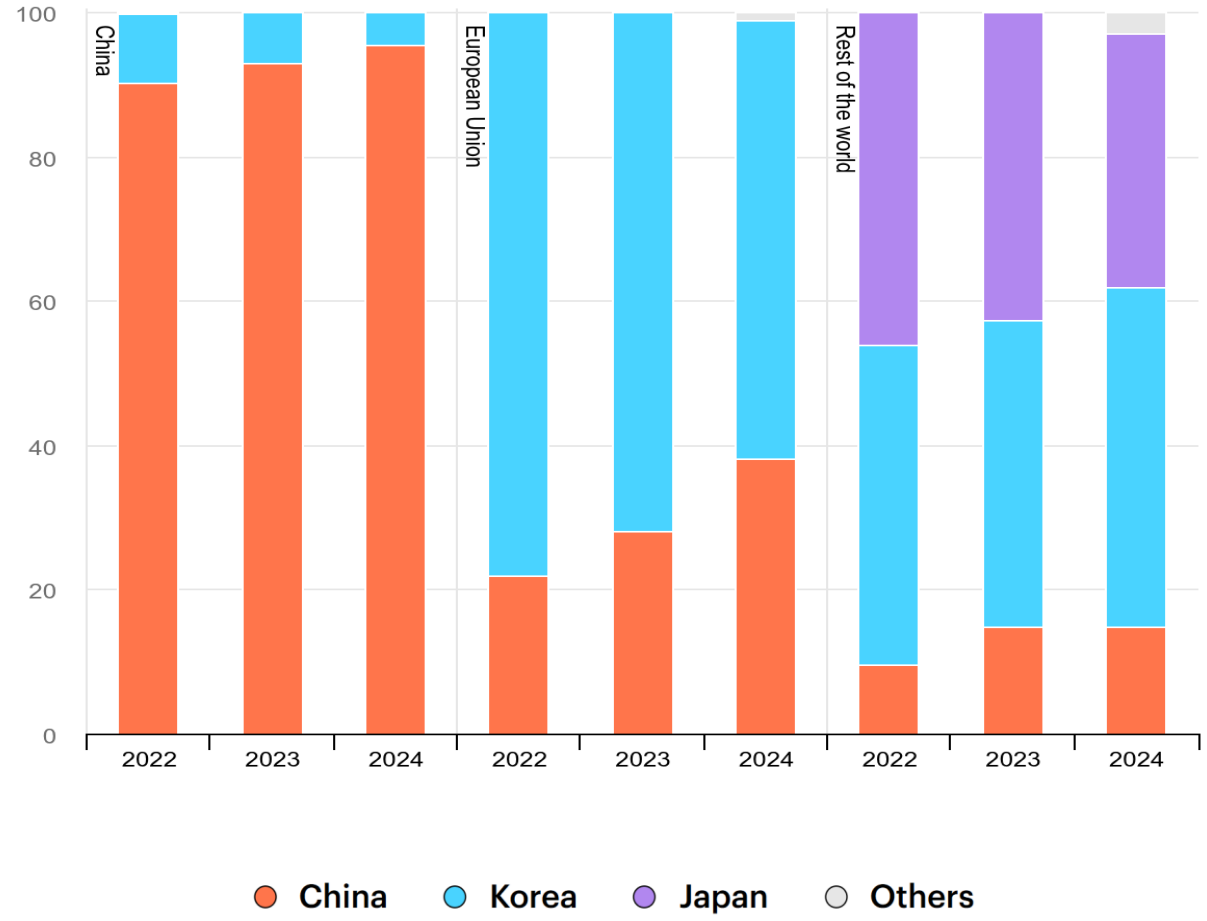
Regional concentration: Today concentration remains a defining characteristic of the global battery industry. In 2025, China accounted for approximately 80% of global battery cell production, highlighting the dominance of Asia in battery manufacturing. This high level of geographic concentration increases supply chain vulnerability to geopolitical tensions, trade disruptions, and other external shocks. Although battery manufacturing capacity is expanding rapidly across North America, Europe, and other regions, many announced projects are still under development. Ensuring that manufacturing capacity grows in line with future demand, while avoiding shortages or excess capacity, remains a key policy and industry challenge.

5.4 China's Dominance and the Role of Korea and Japan

China's primacy is overwhelming. In 2025, Chinese companies (CATL, BYD, CALB, etc.) accounted for roughly **80–85%** of global cell output. China's battery leadership has been supported by sustained industrial policy, including historical purchase subsidies, continuing tax incentives, public funding for battery and new-energy vehicle technology development, and

[6] https://www.iea.org/reports/global-energy-review-2026/technology-battery-storage
[7] https://www.iea.org/reports/batteries-and-secure-energy-transitions
[8] https://www.iea.org/reports/electricity-2026/executive-summary

regulatory instruments such as the dual-credit system. These measures helped create the world's largest domestic EV market and enabled the development of a highly integrated industrial ecosystem spanning overseas mineral investments and sourcing, domestic mineral refining and active-material production, battery-cell manufacturing and electric-vehicle production. Although direct national EV-purchase subsidies were phased out in 2022, tax incentives and wider industrial-policy support continue to reinforce the sector's scale and competitiveness[9].



The figure demonstrates that battery manufacturing capacity remains highly concentrated among companies headquartered in **China, Korea, and Japan**. Chinese firms dominate several battery categories, accounting for more than **90%** of manufacturing capacity in some segments. Korean companies lead in other battery segments, while Japanese manufacturers hold a substantial share in selected technologies. Companies headquartered outside these three countries contribute only a minor proportion of global manufacturing capacity, highlighting the continued regional concentration of the battery industry.

[9] https://english.www.gov.cn/policies/latestreleases/202011/02/content_WS5f9ff225c6d0f7257693ece2.html, https://www.nea.gov.cn/2012-10/16/c_131908404.htm, https://jjs.mof.gov.cn/zhengcefagui/201504/t20150429_1224515.htm, https://english.www.gov.cn/state_council/ministries/2019/07/05/content_281476748683594.+-htm

5.5 Major Global Battery Companies and Business Models

Key manufacturers: The battery sector is dominated by a handful of big players. Leading *cell producers* include: CATL (China's largest, tied with Tesla/BYD among top), BYD (China's largest EV maker with in-house cells), Panasonic (Japan, Tesla partner), LG Energy Solution (Korea), Samsung SDI (Korea), and SK On (Korea). Recent rankings (2025) The global EV battery market in 2025 remained highly concentrated among a few manufacturers, with Chinese companies maintaining a dominant position. CATL led the market with 464.7 GWh of battery installations, accounting for 39.2% of the global market, followed by BYD with 194.8 GWh (16.4%). Together, these two Chinese manufacturers represented more than 55% of global EV battery installations. South Korea's LG Energy Solution ranked third with 108.8 GWh (9.2%), while CALB and Gotion High-Tech, both headquartered in China, held market shares of 5.3% and 4.5%, respectively. Other major manufacturers included SK On, Panasonic, Samsung SDI, EVE Energy, and Svolt, each accounting for between 2.4% and 3.7% of the market. Overall, Chinese manufacturers occupied six of the top ten positions and collectively accounted for approximately 70% of global EV battery installations, highlighting China's continued dominance in the global battery industry. CATL, maintained an 18% operating margin in 2025 (up from a 10-15% range in prior years) and grew overseas revenue to around 35% of its total. By contrast, Korean producers LGES and Panasonic Energy's US operations have been reliant on US production tax credits to stay profitable at all — without them, LGES's 2025 operating result would have been roughly negative USD 200 million rather than positive.[10]

Business models:

- *Integrated EV OEMs:* BYD and Tesla combine vehicle manufacturing with varying degrees of in-house battery production to improve supply security, reduce costs and retain control over battery design. BYD manufactures its LFP-based Blade Battery through FinDreams Battery and supplies both its own vehicles and selected external automakers.[11] Tesla produces its proprietary 4680 cells but continues to procure substantial volumes from specialist suppliers. While integration can create switching costs around particular cell formats and chemistries, both companies retain the ability to diversify their battery technologies and supply arrangements[12].
- *Cell specialists:* CATL, LG, Samsung, SK On, Panasonic, etc., produce cells for many automakers and grid projects. They compete on scale, technology (energy density, fast charging) and incremental innovation. These firms often form long-term supply contracts with OEMs (e.g. LG/GM Ultium Cells, SK/VW). Companies such as CATL, LG Energy Solution, Samsung SDI, SK On and Panasonic Energy supply batteries to multiple automotive and, in several cases, stationary-storage customers. They compete through

[10] https://www.iea.org/reports/global-ev-outlook-2026/manufacturing-and-trade
[11] https://en.byd.com/search/BYD%20batteries
[12] https://www.teslaacessories.com/ro/blogs/news/the-4680-milestone-why-tesla's-in-house-cell-production-changes-everything?srsltid=AfmBOoq6fNrCqWUfhYeioqBM7coI6-0Pz6yX1PMDOiVsBfTnImNAayzM

manufacturing scale, production yield, cost, safety, energy density, charging performance and incremental improvements in cell chemistry and design. Long-term supply agreements and joint ventures increasingly link these manufacturers with automakers, as illustrated by GM–LG Energy Solution's Ultium Cells, Ford–SK On's BlueOval SK and the CATL–Stellantis LFP manufacturing venture in Spain.[13]

- *Upstream and midstream integration*: Companies across the battery value chain are increasingly expanding beyond their traditional activities. Tesla has pursued mineral-supply agreements and is developing in-house lithium refining, cathode-material and cell-production capabilities in the United States. However, its agreement with Piedmont Lithium is an offtake arrangement rather than a mining joint venture. Ganfeng Lithium represents a more extensive example of vertical integration, with activities spanning lithium-resource development, chemical refining, battery manufacturing and recycling.[14][15]
- *Recycling, second-life and service models*: New business models are emerging around battery collection, recycling, material recovery, repurposing and subscription-based access. Redwood Materials is developing a closed-loop model that connects battery recycling with the production of battery-grade materials, while Sacyr's Baterurgia consortium is developing technologies for dismantling and recovering strategic materials[16]. Li-Cycle should now be identified as part of Glencore Battery Recycling following its 2025 acquisition[17]. For second-life applications, BMW and Mercedes-Benz provide established examples of retired vehicle batteries being used in stationary storage systems[18]. Battery-as-a-Service is most clearly demonstrated by NIO's battery-subscription and swapping model[19], while Renault has previously offered battery leasing with capacity guarantees.[20]

5.6 Battery Cost Trends, Price Volatility and Supply Chain Risks

Cost trends: Lithium-ion battery costs have fallen sharply because of manufacturing scale, technological learning, intense competition and lower raw-material prices. Battery-pack prices declined by 20% in 2024, the largest annual fall since 2017, and BloombergNEF reported a further reduction to an average of $108 per kWh in 2025. This compares with $139 per kWh in 2023. Lithium prices also returned close to their late-2015 level in 2024, even though global lithium

[13] https://www.iea.org/reports/global-ev-outlook-2026/electric-vehicle-batteries
[14] https://www.sec.gov/Archives/edgar/data/1728205/000172820520000031/ex99_1.htm, https://www.tesla.com/blog/tesla-lithium-refinery-groundbreaking, https://www.ganfenglithium.com/index_en.html
[15] https://sacyr.com/en/-/proyecto-innovacion_reciclaje-baterias
[16] https://sacyr.com/en/-/reciclamos-y-valorizamos-las-baterias-de-litio-de-coches-electricos/blog
[17] https://www.glencore.com/what-we-do/recycling
[18] https://group.mercedes-benz.com/sustainability/resources-circularity/recycling/battery.html, https://www.press.bmwgroup.com/global/article/detail/T0275547EN/bmw-group-underlines-leading-role-in-electro-mobility?language=en
[19] https://www.nio.com/news/nio-launches-battery-service
[20] https://www.renaultgroup.com/en/magazine/energy-and-powertrains/all-you-need-to-know-about-battery-leasing-for-the-renault-zoe

demand had increased approximately sixfold over the same period. Further cost reductions are expected as manufacturers expand production, improve factory efficiency, adopt simpler pack designs and increase the use of lower-cost LFP batteries. [21]

The commercial importance of the industry is nevertheless expanding. McKinsey estimated that battery value-chain revenues could increase from approximately $85 billion in 2022 to more than $400 billion by 2030. Active-material production and battery-cell manufacturing are expected to represent the largest revenue opportunities.[22]

Raw-material price volatility: Demand for key energy minerals continued to grow strongly in 2024. Lithium demand rose by nearly 30%, significantly exceeding the 10% annual growth rate seen in the 2010s. Demand for nickel, cobalt, graphite and rare earths increased by 6-8% in 2024. This growth was largely driven by energy applications such as electric vehicles, battery storage, renewables and grid networks. In the case of copper, the rapid expansion of grid investments in China has been the single largest contributor to demand growth over the past two years. For battery metals such as lithium, nickel, cobalt and graphite, the energy sector accounted for 85% of total demand growth over the same period.[23]

Despite this rapid demand growth, major supply increases – led by China, Indonesia and the Democratic Republic of the Congo – exerted downward pressure on prices, especially for battery metals. The swift increase in battery metal production highlighted the sector's ability to scale up new supply more quickly than for traditional metals like copper and zinc. Since 2020, supply growth for battery metals has been twice the rate seen in the late 2010s. As a result, following the sharp price surges of 2021 and 2022, prices for key energy minerals have continued to decline, returning to pre-pandemic levels. Lithium prices, which had surged eightfold during 2021-22, fell by over 80% since 2023. Graphite, cobalt and nickel prices also dropped by 10 to 20% in 2024.

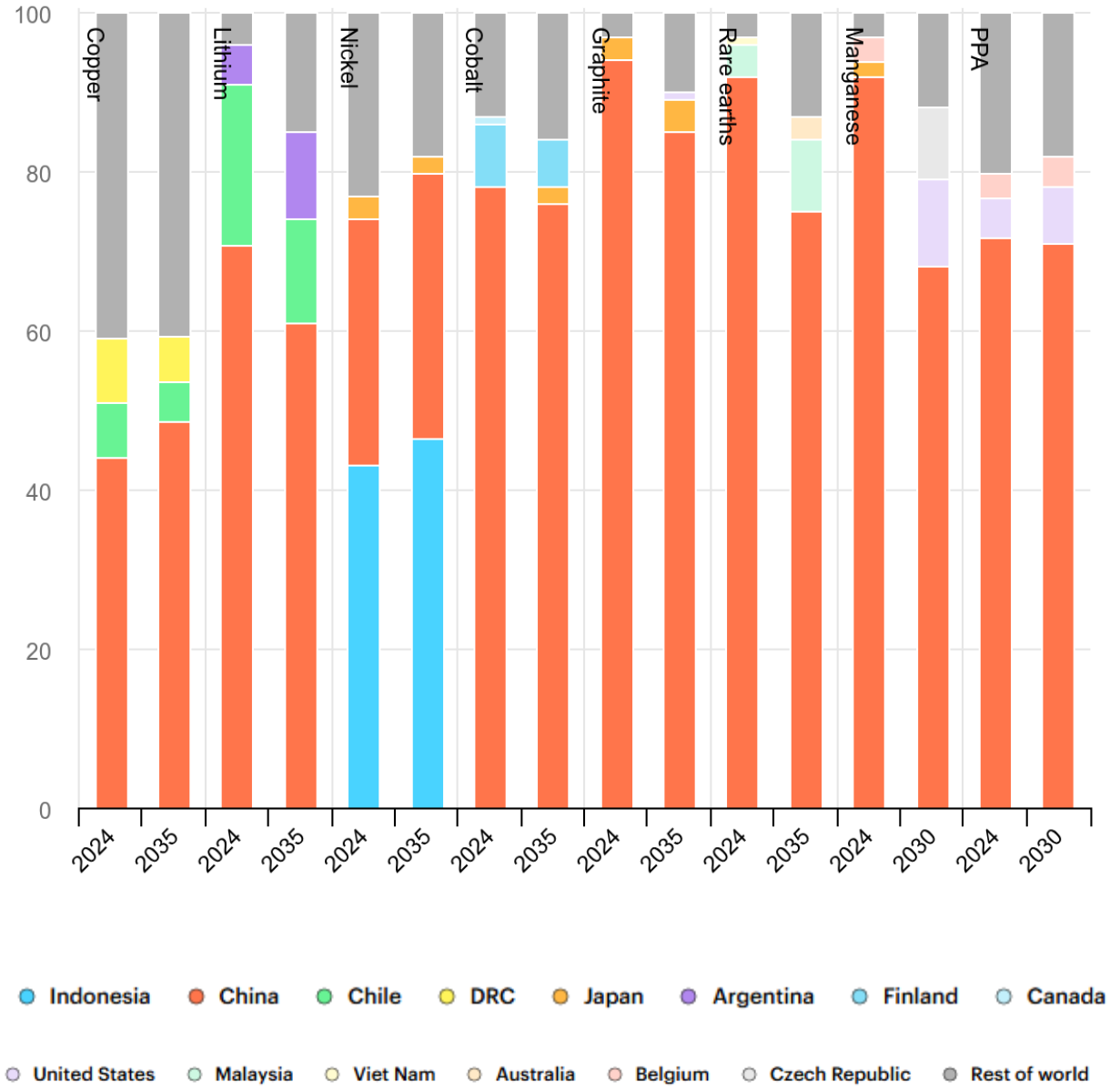
Supply-chain concentration: Based on the project pipeline, the geographical concentration of mining operations is set to remain high for most minerals. Mined supply is set to increase in concentration in the top three countries for copper, nickel and cobalt by 2035. In 2035, the top three producing countries for nickel supply 85% of the market, up from 75% in 2024. Indonesia is set to see significant supply growth in both nickel and cobalt markets over the next decade. There is some diversification emerging in the mining of lithium, graphite and rare earth elements. The share of mined lithium supply from the top three producers is set to fall below 70% by 2035, down from over 75% in 2024. Graphite and rare earth elements also see some improvement as new

[21] https://www.iea.org/reports/global-ev-outlook-2025/electric-vehicle-batteries, https://about.bnef.com/insights/clean-transport/lithium-ion-battery-pack-prices-fall-to-108-per-kilowatt-hour-despite-rising-metal-prices-bloombergnef/, https://about.bnef.com/insights/clean-energy/lithium-ion-battery-pack-prices-hit-record-low-of-139-kwh

[22] https://www.mckinsey.com/industries/automotive-and-assembly/our-insights/battery-2030-resilient-sustainable-and-circular

[23] https://www.iea.org/reports/global-critical-minerals-outlook-2025/executive-summary

mining suppliers emerge over the next decade – Madagascar and Mozambique for graphite and Australia for rare earths.



However, refining operations for most minerals are set to remain highly concentrated over the next decade. Refining concentration increases significantly for nickel due to major growth in supply in Indonesia. Despite some diversification occurring for lithium and others, China remains the dominant refined supplier for almost all minerals. In 2035, China is set to supply over 60% of refined lithium and cobalt, and around 80% of battery-grade graphite and rare earth elements.[24]

[24] https://www.iea.org/reports/global-critical-minerals-outlook-2025/overview-of-outlook-for-key-minerals

6. End-of-Life Battery Recycling and Circular Economy

6.1 Why End-of-Life Battery Management Matters

The rapid expansion of electric mobility, consumer electronics and battery energy storage systems is creating a future stream of batteries that will eventually require structured collection, assessment, reuse and recycling. In the near term, manufacturing scrap remains the largest source of lithium-ion recycling feedstock. However, the International Energy Agency expects end-of-life electric-vehicle and storage batteries to become the largest source from around 2035 and to account for more than 90% of available battery recycling feedstock by 2050.[25]

The central challenge is that a battery reaching the end of its first application may still contain two different forms of value:

1. **Residual energy value**, where the battery can continue operating in the same or a less demanding application; and
2. **Material value**, where lithium, nickel, cobalt, manganese, copper, aluminium, graphite, lead and other materials can be recovered.

An effective end-of-life system must therefore avoid two opposite errors: sending usable batteries for premature recycling, and retaining unsafe or technically exhausted batteries in informal reuse markets.

Dimension	**Strategic relevance**
Environmental protection	Prevents uncontrolled dumping, leakage of electrolytes, unsafe burning and contamination of soil, air and water
Public safety	Reduces risks from short circuits, thermal runaway, fire and unsafe dismantling
Critical mineral security	Recovers lithium, nickel, cobalt, graphite, copper, manganese and other materials from domestic waste streams
Economic value	Converts used batteries and production scrap into second-life products, black mass and refined materials
Industrial resilience	Creates a secondary material supply for battery and component manufacturing
Regulatory compliance	Enables producers to meet collection, recycling, refurbishment and material-recovery obligations

[25] https://www.iea.org/reports/recycling-of-critical-minerals/executive-summary

Dimension	Strategic relevance
Climate impact	Reduces part of the energy use and emissions associated with primary mineral extraction and refining

It is estimated that a successful global scale-up of recycling could reduce the need for new mining activity by approximately 25–40% for selected energy-transition minerals by 2050. It also estimates that recycled lithium, nickel and cobalt can, on average, involve around 80% lower greenhouse-gas emissions than production from primary mining and refining. Recycling will not eliminate the need for new mines, but it can reduce future dependence on primary extraction and provide a more resilient secondary source of materials.[26]

For India, end-of-life management is also a legal obligation. The Battery Waste Management Rules, 2022 place Extended Producer Responsibility on producers, require collected waste batteries to be sent for recycling or refurbishment, and prohibit their disposal through landfilling or incineration. The Rules also permit mechanisms such as buy-back and deposit-refund systems to create dedicated collection streams.[27]

6.2 Battery Lifecycle and End-of-Life Pathways

A battery's lifecycle begins with raw-material extraction and processing, followed by component and cell manufacturing, module and pack assembly, integration into an application, first-life use and eventual retirement. Retirement from the first application does not automatically mean that the battery has become waste. Its next pathway depends on safety, physical condition, chemistry, performance history, residual capacity, future application requirements and the economics of testing and reconfiguration.

The attached Punjab report presents the battery circularity chain as:

Collection → Testing and Grading → Dismantling → Repurposing → Material Recovery → Manufacturing Linkage

This is useful because it positions recycling as part of a closed industrial loop rather than as a standalone waste-treatment activity.

Pathway	Meaning	Typical condition	Possible outcome
Direct reuse	Battery continues in the same application with limited intervention	Battery remains safe and performs within required limits	Reinstallation, resale or continued fleet use

[26] https://www.iea.org/reports/recycling-of-critical-minerals/executive-summary
[27] https://gemrecycling.com/wp-content/uploads/2023/04/Battery-Waste-Mgmnt-Rules-2022.pdf

Refurbishment	Faulty components, modules, wiring, casing or control systems are repaired or replaced	Battery performance remains adequate but specific components have failed	Return to original or similar application
Repurposing	Battery is redesigned for a less demanding application	Battery is no longer suitable for its original use but retains adequate functionality	Stationary storage, backup power, microgrids or charging support
Recycling	Battery is dismantled and processed to recover materials	Battery is technically exhausted, unsafe, damaged or uneconomic to reuse	Black mass, metal salts, alloys, copper, aluminium, plastics and other outputs
Residual treatment	Non-recoverable process residues are managed under applicable hazardous-waste requirements	Material cannot be safely or economically recovered	Controlled treatment and disposal of process residues

The United Nations Development Programme similarly distinguishes refurbishment, repurposing and recycling as separate post-use pathways. Batteries with repairable pack-level faults may return to their original use; those unsuitable for mobility may be redesigned for stationary storage; and those that cannot be safely reused are sent for material recovery. [28]

6.3 Collection, Testing, Grading and Second-Life Applications

Collection and reverse logistics: Collection is the foundation of the end-of-life ecosystem because it determines the quantity, quality and predictability of feedstock reaching formal operators. Waste batteries arise from dispersed sources including households, dealerships, service centres, vehicle manufacturers, fleet operators, battery-swapping stations, telecom facilities, industrial users, energy-storage projects and accident-damaged vehicles.

Unlike conventional metal scrap, lithium-ion batteries require controlled handling during storage and transportation. Damaged packs may retain electrical charge and may present fire or chemical risks. Batteries must therefore be segregated by chemistry and condition, protected from short circuits, appropriately packaged and transferred through traceable channels.

Under India's Battery Waste Management Rules, entities involved in collection and segregation are required to hand waste batteries over to registered refurbishers or recyclers. Producers may

[28] https://www.undp.org/sites/g/files/zskgke326/files/2025-01/analysis-of-ev-battery-end-of-life.pdf

establish buy-back, deposit-refund or other collection models, although the responsibility for meeting Extended Producer Responsibility targets remains with the producer.

Local collection points → District aggregation facilities → Regional testing and grading centres → Repurposing or recycling facilities

This hub-and-spoke structure reduces the cost of moving small, scattered quantities directly to distant processing plants and creates an intermediate point for safe storage, documentation and consolidation.

Testing and grading: Testing and grading form the decision-making core of the circular battery system. The objective is to determine whether a battery should be reused, refurbished, repurposed or recycled.

Testing stage	Indicative assessment
Identification	Manufacturer, chemistry, model, serial number, ownership and application history
Visual inspection	Swelling, leakage, corrosion, deformation, impact damage and connector condition
Electrical safety	Isolation, voltage, insulation resistance and short-circuit risk
BMS assessment	Error codes, temperature history, cycle history, cell imbalance and fault records
Performance assessment	State of Charge, State of Health, internal resistance and remaining capacity
Module/cell testing	Identification of weak or damaged modules and consistency across cells
Application matching	Whether the battery can safely meet the power, energy and cycle requirements of a second use

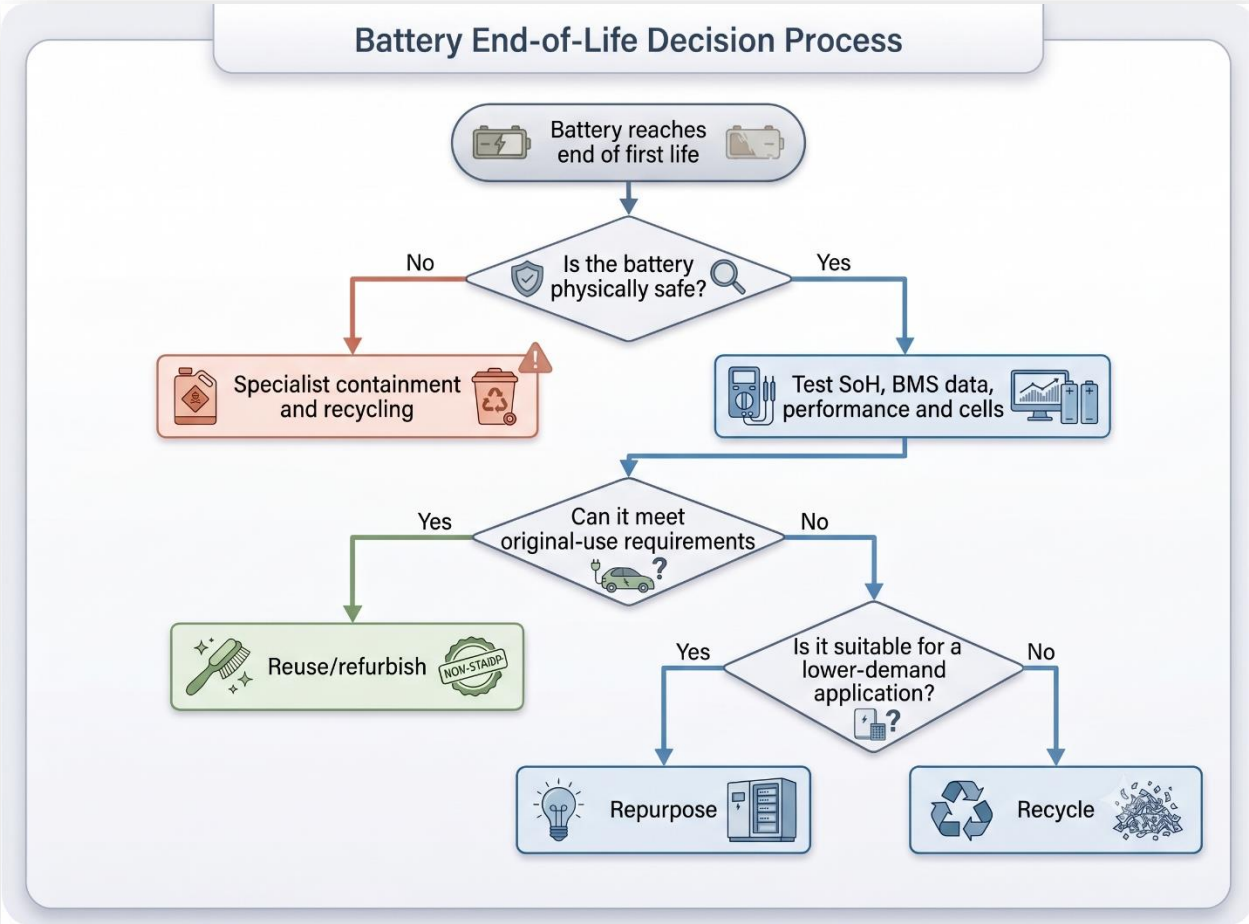
A structured assessment should follow this process

India currently lacks a unified national testing and certification protocol for second-life batteries. A Centre for Science and Environment assessment notes that repurposing entities often need to validate each pack independently; conventional testing can require four to five hours per pack, increasing cost and restricting throughput. Access to standardised Battery Management System data and digital diagnostic tools could significantly shorten this process and improve safety.

Second-life applications: Retired mobility batteries may be suitable for applications with lower power demands, more predictable operating conditions or less stringent weight and space constraints. Potential uses include:

- Telecom Backup;
- Solar And Wind Energy Storage;
- Rural Or Commercial Microgrids;
- EV Charging-Station Buffer Storage;
- Residential And Commercial Backup;
- Industrial Peak-Load Management; And
- Selected Battery Energy Storage System Applications.

Second life can delay recycling and extract more service from the energy and materials already embedded in the battery. However, viability depends on testing costs, reconfiguration costs, warranty conditions, access to BMS data, performance consistency, fire-safety systems and the value of an equivalent new battery.



6.4 Battery Recycling Technologies and Black Mass Recovery

Battery recycling routes differ by chemistry. Lead-acid batteries operate through a relatively mature closed-loop system because their design is standardised, material value is clear and collection networks are well established. Lithium-ion batteries are more complex due to variation in pack design, chemistry and material composition.

Lead-Acid Recycling

A typical lead-acid recycling process follows:

Collection → Battery breaking → Separation of lead, plastic and electrolyte → Lead refining → Reuse in new batteries

This model demonstrates how standardised products, take-back systems and strong regulation can support high recycling rates. However, informal smelting and unsafe acid handling can create serious environmental and health risks.[29]

Lithium-Ion Recycling and Black Mass

Lithium-ion recycling generally involves two stages:

1. **Pre-treatment:** collection, safe storage, discharge, dismantling, shredding and separation of metals, plastics and other components.
2. **Material recovery:** extraction of valuable materials through pyrometallurgy, hydrometallurgy, direct recycling or hybrid processes.

The main intermediate output is **black mass**, a fine material containing lithium, graphite, nickel, cobalt, manganese and other compounds depending on battery chemistry. Black mass is not the final recycled product; it must be further refined into usable metal salts or regenerated battery materials.

Comparison of Recycling Technologies

Technology	Key Strength	Key Limitation	Status
Pyrometallurgy	Handles mixed and contaminated feedstock	High energy use; lower recovery of lithium and graphite	Commercially established
Hydrometallurgy	High recovery potential for lithium, nickel, cobalt and manganese	Requires chemical and wastewater management	Commercially expanding

[29] https://www.iea.org/reports/recycling-of-critical-minerals/executive-summary

Direct recycling	Preserves cathode material value and may reduce processing intensity	Requires clean, chemistry-specific feedstock	Emerging
Hybrid process	Combines feedstock flexibility with higher recovery	More complex plant design	Increasingly common

Hydrometallurgy and pyrometallurgy remain the main commercial routes, while direct recycling is still developing. Technology choice depends on chemistry, feedstock quality, scale, recovery targets and environmental controls.[30]

Chemistry and Recycling Economics

Nickel- and cobalt-rich batteries generally provide higher recycling value than lithium iron phosphate batteries. LFP contains lithium, copper, aluminium and graphite but lacks high-value nickel and cobalt. As LFP use grows, recycling business models may increasingly depend on processing fees, EPR obligations, recycled-content demand and efficient recovery technologies rather than commodity value alone.[31]

6.5 Economics, Safety and Compliance Challenges

Recycling economics

The financial viability of battery recycling is determined by the interaction of several factors rather than plant capacity alone.

Driver	Influence on profitability
Battery chemistry	Determines the value and concentration of recoverable materials
Feedstock volume	Influences plant utilisation and economies of scale
Feedstock quality	Mixed or contaminated feedstock increases sorting and refining costs
Collection distance	Affects reverse-logistics and hazardous-transport costs
Pack design	Determines the labour and automation required for disassembly
Recovery yield	Higher material recovery increases output value
Product purity	Battery-grade products generally command more value than low-grade intermediates
Commodity prices	Lithium, nickel and cobalt price movements directly affect revenues

[30] https://wri-india.org/perspectives/decoding-economic-viability-lithium-ion-battery-recycling
[31] https://www.iea.org/reports/recycling-of-critical-minerals/executive-summary

Driver	Influence on profitability
Energy and chemicals	Major operating costs for thermal and hydrometallurgical processes
Offtake agreements	Reduce revenue uncertainty and support financing
Compliance cost	Includes registration, reporting, environmental controls, storage and worker safety

Table: Principal drivers of recycling economics

Recycling economics indicates that infrastructure, equipment and installation form a significant share of project capital costs, while chemicals, utilities and labour are major operating costs. It also finds that NMC and NCA batteries generally offer stronger material revenues than LFP and LMO, and that the most viable recycling technology varies by chemistry and scale.[32]

This creates a central investment risk: a recycler may build capacity based on expected future EV retirements but face low utilisation in the near term. Feedstock agreements with manufacturers, electronics companies, fleet operators and storage developers are therefore as important as the processing technology.

Safety challenges: Battery recycling involves interconnected electrical, thermal and chemical hazards.

Stage	Principal risks
Collection	Unidentified chemistry, damaged batteries, exposed terminals and mixed storage
Transport	Short circuit, movement damage, fire and inadequate packaging
Storage	Thermal runaway propagation and accumulation of damaged batteries
Discharging	Electric shock, overheating and uncontrolled energy release
Dismantling	Sharp components, live circuits, electrolyte exposure and fire
Shredding	Dust, residual charge, volatile compounds and ignition
Hydrometallurgy	Acid handling, chemical exposure, wastewater and reaction control
Pyrometallurgy	High temperature, off-gases, slag and energy intensity

[32] https://wri-india.org/perspectives/decoding-economic-viability-lithium-ion-battery-recycling

Stage	Principal risks
Black-mass handling	Fine particulate exposure, contamination, fire and unauthorised movement

Safety therefore needs to be designed across the entire reverse supply chain, not added only at the recycling plant. Separate storage zones, battery isolation, temperature monitoring, fire detection and suppression, emergency-response procedures, trained personnel and appropriate packaging are essential.

Compliance framework in India: India's Battery Waste Management Rules establish a national EPR framework covering producers, recyclers and refurbishers. Key requirements include:

- Producer registration and submission of EPR plans;
- Collection and recycling or refurbishment obligations;
- Registration of recyclers and refurbishes;
- Category-wise processing certificates;
- Centralised portal-based reporting;
- Periodic returns;
- Material-recovery requirements;
- Inspection and audits; and
- Environmental compensation for non-compliance.

The Rules also define refurbishment as repairing, reconditioning or repurposing a used battery for second life. Collected waste batteries must be sent to registered recycling or refurbishment entities and cannot be landfilled or incinerated.

Key compliance weakness: The existence of certificates and recovery targets does not by itself prove that batteries have physically moved through the formal chain or that high-quality materials have been recovered. A credible system must reconcile:

Battery placed on market → Battery collected → Battery received by authorised operator → Battery processed → Material recovered → Recovered material sold or reused

Without physical verification and material-balance reconciliation, EPR risks becoming a certificate market disconnected from actual battery flows.

6.6 Informal Sector Risks and Formalisation Needs

Integrating the Informal Collection Ecosystem into the Circular Economy

A resilient circular economy for batteries depends not only on recycling capacity but also on an efficient and inclusive collection ecosystem. In India, informal battery collectors and aggregators play a significant role in recovering end-of-life batteries. Their extensive local networks, rapid

collection mechanisms, and ability to reach dispersed generators enable them to recover batteries from areas that are often beyond the reach of formal collection systems.

However, much of this material currently moves through channels that operate outside formal regulatory and reporting frameworks. As a result, valuable battery resources are not always traceable throughout the value chain, limiting opportunities for safe handling, material recovery, and effective implementation of Extended Producer Responsibility (EPR). Strengthening material traceability and channeling end-of-life batteries into authorised recycling systems is therefore an important aspect of building a more circular battery value chain.

Rather than viewing the informal sector solely as a challenge, it can be recognised as an existing collection infrastructure with significant operational knowledge and market reach. An inclusive circular economy approach seeks to build on these strengths by creating pathways for integration into the formal system. Measures such as tiered registration, simplified compliance requirements, capacity building, financial incentives, and partnerships with authorised recyclers can enable informal collectors to participate in regulated value chains while preserving their livelihoods.

By integrating informal collection networks into the formal ecosystem, policymakers can improve collection rates, enhance material traceability, strengthen resource recovery, and create a more efficient and socially inclusive circular economy for end-of-life batteries.

7. India's Battery Ecosystem: Manufacturing Readiness, Policy Landscape and Strategic Opportunities

India's battery demand is entering a phase of structural expansion, driven by the simultaneous electrification of transport, rapid growth in renewable energy, increasing deployment of battery energy storage systems (BESS), expansion of digital infrastructure, and continued demand from consumer electronics. Unlike many mature battery markets where electric vehicles dominate demand, India's battery ecosystem is expected to evolve through multiple demand centres simultaneously. Electric mobility remains the largest growth engine, but stationary storage, telecom backup, data centres and distributed renewable-energy systems are becoming increasingly important as the country pursues energy security and grid reliability. This broad-based demand profile creates opportunities across the battery value chain while also increasing the urgency of developing domestic manufacturing, recycling and critical-mineral supply chains.

7.1 India’s Battery Demand Landscape

- **Rapid EV growth:** Electric vehicle sales rose from ~50,000 in 2016 to 2.08 million in 2024, driven largely by two-, three-, and four-wheelers. Current EV penetration ~8.5% in FY2025-26.[33] While passenger electric cars attract considerable public attention, India's battery demand is presently dominated by electric two-wheelers and three-wheelers because of their lower

[33] https://jmkresearch.com/indias-ev-penetration-reached-8-5-in-fy2025-26/

acquisition costs, high daily utilization, and favourable total cost of ownership. Electrification of buses and commercial vehicles is also gaining momentum through government procurement programme and state transport initiatives. Consequently, India's battery market differs from that of Europe or North America, where passenger cars account for a much larger share of battery demand. This difference has important implications for battery chemistry selection, manufacturing priorities and charging infrastructure requirements.[34]

	E-2W	E-3W P	E-3W C	E- Cars	E- Bus	E-GV	Total
FY2025-26	6.6%	63.1%	52.2%	4.4%	4.4%	1.4%	8.5%
FY2024-25	6.2%	59%	49.6%	2.6%	3.6%	0.6%	7.7%

- **Battery demand scale & Future projections**: India's battery demand is expected to increase dramatically for (EV + stationary + consumer) under an accelerated adoption scenario, rising from approximately 15 GWh in 2022 to nearly 260 GWh by 2030. This growth is primarily driven by rapid electrification of the transport sector, particularly passenger and commercial electric vehicles, alongside expanding deployment of grid-scale battery energy storage systems. Consequently, the domestic battery market is projected to grow from US$2 billion to around US$15 billion over the same period. The findings highlight the urgent need to strengthen domestic battery manufacturing, develop recycling infrastructure, and establish a secure circular battery value chain to meet future demand.[35]

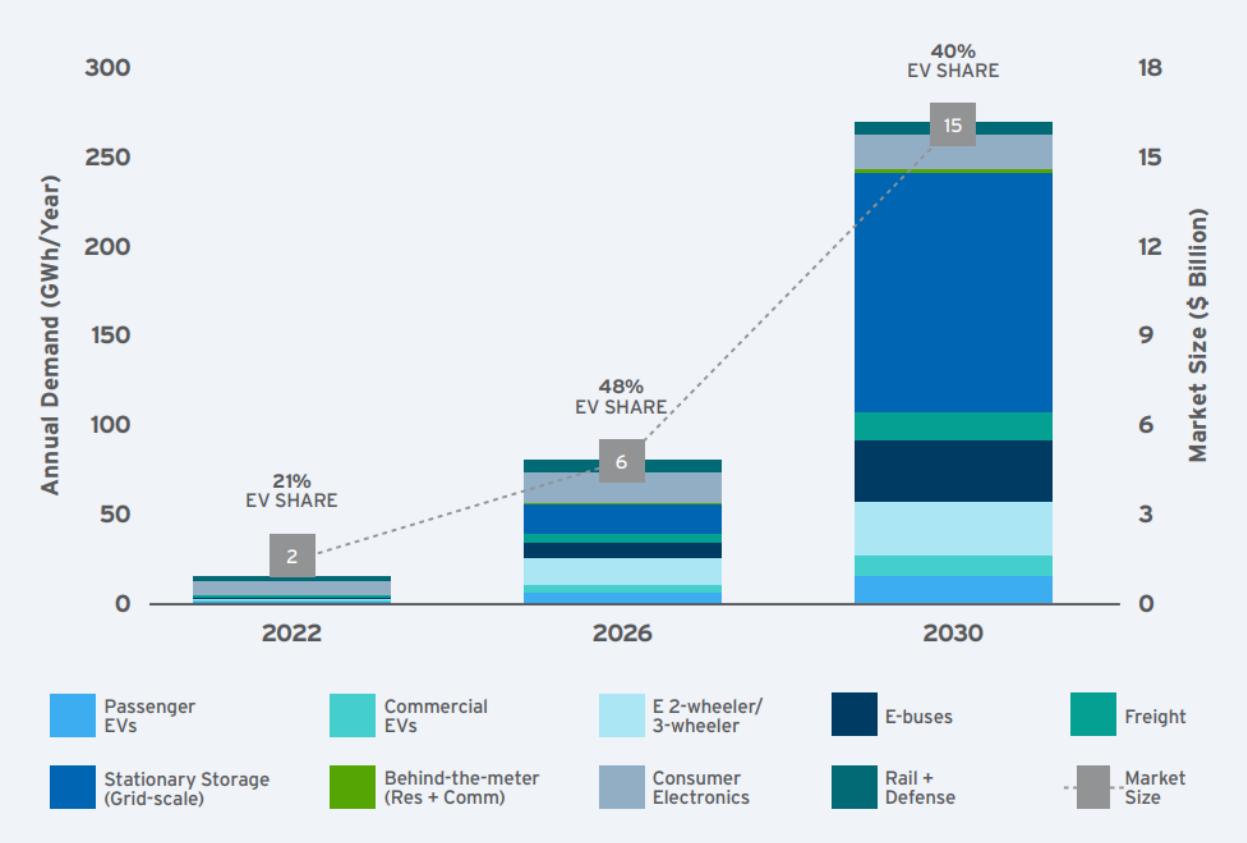
- **Tri-angulation with two independent estimates:** WRI India projects demand rising from 20 GWh (2022) to 220 GWh by 2030 (50% CAGR), reaching 1,067 GWh by 2047. IEEFA/JMK Research project a 36.5% CAGR scaling ACC demand to ~272 GWh by FY2030; the India Energy Storage Alliance (IESA) forecasts 700+ GWh by the mid-2040s.[3637]

- **All three estimates (NITI Aayog 260 GWh, WRI's 220 GWh, IEEFA/JMK's 272 GWh)** cluster in the same range — worth stating as a triangulated range (roughly 220-270 GWh by 2030) rather than a single number. Although individual projections differ because they use different assumptions regarding EV adoption, battery energy storage deployment and industrial electrification, all major studies point to the same conclusion: India's advanced chemistry cell (ACC) demand is expected to increase by roughly an order of magnitude before 2030. Rather than relying on a single forecast, it is therefore more appropriate to interpret these estimates as a projected demand range of approximately 220–270 GWh by 2030, indicating strong

[34] https://www.niti.gov.in/sites/default/files/2025-08/India-Electric-Mobility-Index-2024-Report.pdf
[35] https://www.niti.gov.in/sites/default/files/2023-03/Need-for-ACC-Energy-Storage-in-India.pdf
[36] https://wri-india.org/sites/default/files/2026-01/India%E2%80%99s%20battery%20ecosystem.pdf
[37] https://ieefa.org/resources/securing-indias-battery-supply-chain-more-critical-ever

consensus regarding the scale of future market expansion despite uncertainty over the exact trajectory.

- **Current-year data point:** India's actual ACC battery demand in 2025 was 28 GWh, split ~60% EV and ~40% stationary/grid storage — useful because it shows storage is already a much larger share of near-term demand than EV-focused narratives suggest.[36] The current demand mix also illustrates that India's battery ecosystem is gradually diversifying. Although electric mobility remains the principal demand driver, stationary energy storage is emerging more rapidly than previously anticipated because of increasing renewable-energy integration, utility-scale storage tenders and growing electricity demand from commercial and digital infrastructure. This suggests that future battery policy cannot remain exclusively EV-centric and must increasingly consider stationary storage as an equally important growth market.



Expected Growth in Indian Battery Demand (Accelerated Scenario)[1]

- **Demand-driver imbalance:** The rapid expansion of battery demand has exposed a widening gap between India's consumption requirements and its domestic manufacturing capabilities. Despite the launch of the Advanced Chemistry Cell Production Linked Incentive (ACC PLI) scheme to establish 50 GWh of domestic cell manufacturing capacity, implementation has progressed more slowly than anticipated. As of October 2025, only 1.4 GWh of the targeted

capacity had been commissioned, with no incentives disbursed because beneficiaries had yet to meet the required domestic value-addition criteria. Progress has been constrained by delays in project execution, continued dependence on imported battery materials and manufacturing equipment, the absence of a mature upstream ecosystem, and shortages of specialised technical expertise. Consequently, India continues to rely on imported lithium-ion cells to meet the growing demand from electric mobility and stationary energy storage, highlighting that strengthening domestic manufacturing will require not only capacity expansion but also the development of an integrated battery value chain spanning critical minerals, cell components, technology, and skilled human resources.[38]

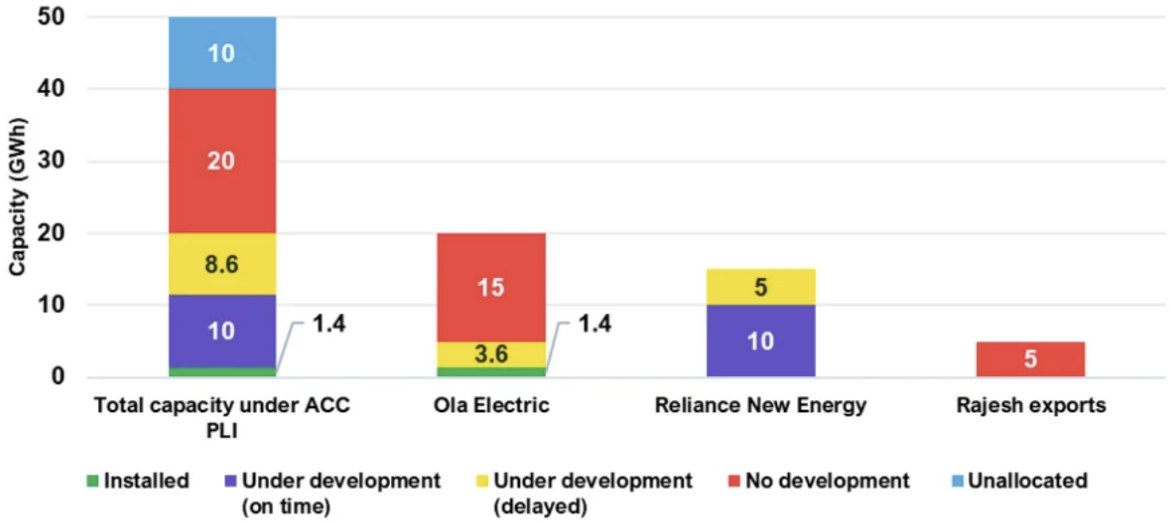
7.2 India's Manufacturing Position: Cells, Packs and BMS

Battery manufacturing is the foundation of India's ambition to build a competitive battery ecosystem. While battery demand is expanding rapidly, value capture depends not only on battery consumption but also on the country's ability to manufacture advanced chemistry cells, battery packs, battery management systems (BMS), and critical battery components domestically. At present, India possesses a growing battery pack assembly industry supported by strong automotive and electronics manufacturing capabilities. However, domestic production of advanced chemistry cells remains at an early stage, with significant dependence on imported cells, active materials, separators, electrolytes and manufacturing equipment. Bridging this manufacturing gap has therefore become a central objective of India's industrial policy.

- **Cell production & Pack assembly:** As of 2023, announced advanced cell (ACC) capacity in India is only ~18 GWh (mostly piloted by PLI-ACC recipients). The PLI-ACC scheme targets 50 GWh by 2026. An additional 10 GWh tranche was tendered in 2024. However, actual commissioning is early-stage and "Ola Cell Technologies Pvt. Ltd" installed their capacity of 1 GWh out of capacity awarded 20 GWh[39] — Exide, Amara Raja, Ola Electric, Uno Minda, Trontek, Amptek . The implementation experience of the ACC PLI scheme demonstrates that financial incentives alone are insufficient to establish a globally competitive battery manufacturing industry. Unlike conventional manufacturing sectors, advanced chemistry cell production depends on an integrated ecosystem comprising specialised machinery, qualified technical personnel, battery-grade raw materials, component suppliers and established quality-control systems. Since much of this ecosystem is still developing in India, several beneficiaries have experienced delays in commissioning production facilities despite strong policy support.[40]

[38] https://www.pib.gov.in/PressReleasePage.aspx?PRID=2246037&reg=48&lang=2
[39] https://www.pib.gov.in/PressReleasePage.aspx?PRID=2246037&reg=48&lang=2
[40] https://jmkresearch.com/renewable-sector-published-reports/assessing-indias-pli-scheme-for-advanced-chemistry-cells-2/

Status of capacity and progress by beneficiaries as of October 2025



Source: News Articles, JMK Research

- **Battery management system**: BMS development is limited. India has an emerging base of domestic BMS developers, but advanced automotive-grade systems continue to depend partly on imported semiconductors, sensing components, software platforms and foreign technology. Localisation is considerably weaker further upstream, with cathode and anode active materials, separators, electrolytes and specialised cell-manufacturing equipment remaining substantially import-dependent, although several domestic pilot and commercial projects are under development.[41]

- **Localisation Gap**: Although India has made measurable progress in downstream battery manufacturing, localisation across the battery value chain remains uneven. Most domestic capabilities are concentrated in battery-pack assembly, system integration and application development, while upstream activities—including critical mineral processing, cathode and anode active material production, separator manufacturing and electrolyte production—remain limited. According to the RMI–Micelio *EVolving Ecosystems* report, India's battery value chain continues to face significant gaps in raw material processing, battery-grade chemical production and component manufacturing, resulting in heavy dependence on imports for both battery materials and manufacturing equipment. Furthermore, the report notes that China currently dominates global battery manufacturing, producing approximately 75% of the world's lithium-ion battery cells, 70% of global cathode capacity, 90% of anode and electrolyte production, and more than half of global lithium, cobalt and graphite refining, highlighting the scale of India's localisation challenge. Consequently, although India is strengthening its downstream manufacturing ecosystem through initiatives such as the ACC PLI Scheme, achieving meaningful domestic value addition will require simultaneous investments in

[41] https://www.niti.gov.in/sites/default/files/2023-07/Niti-Aayog_Report-VS_compressed_compressed.pdf

upstream mineral refining, battery component manufacturing, technology transfer and supply-chain development to reduce import dependence and improve long-term competitiveness.[42]

- **Investment and costs:** Despite the current manufacturing challenges, India's battery sector continues to attract strong investment interest owing to supportive government policies and rapidly growing domestic demand. The ACC PLI Scheme alone was designed to establish **50 GWh** of domestic advanced chemistry cell manufacturing capacity with a financial outlay of **₹18,100 crore (approximately US$2.1 billion)**. Beyond the PLI scheme, several major industrial groups—including Reliance New Energy, Ola Electric, Amara Raja Energy & Mobility, Exide Industries and Tata Group—have announced large-scale investments across cell manufacturing, battery-pack production and battery recycling. However, as highlighted by JMK Research, implementation has progressed more slowly than anticipated, with only **1.4 GWh (2.8%)** of the targeted ACC PLI capacity commissioned by October 2025 and no incentives disbursed because the required domestic value-addition criteria had not yet been achieved. These findings indicate that capital investment alone is insufficient to establish a globally competitive battery industry. Over the medium term, India's competitiveness will depend on developing an integrated manufacturing ecosystem encompassing critical mineral processing, battery component production, specialised technical skills, research and development, and secure international supply partnerships. Successfully addressing these structural gaps would enable India to capture a greater share of value across the global battery supply chain rather than remaining primarily a battery-consuming market.[43][44]

7.3 Critical Mineral Security: Challenges and Strategic Initiatives

India’s electric mobility shift depends not only on the country’s vehicle production capacity and charging infrastructure but also, more fundamentally, on the availability, processing, and financing of critical minerals and battery materials. While lithium-ion batteries require a specific set of materials — lithium, nickel, cobalt, graphite, manganese, copper, aluminium, and iron — the strategic risk for India lies in the uneven domestic availability, processing capacity, and import dependence in key areas of the battery value chain.

This section presents a systems-level evaluation of the critical mineral system in India, considering geological availability, processing and refining capacity, material flow potential recycling, and financing requirements. Rather than focusing on the geological availability of minerals, this evaluation places greater emphasis on the positioning of the value chain and the security of materials.

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[42] https://rmi.org/app/uploads/dlm_uploads/2024/11/evolving_ecosystems.pdf
[43] https://jmkresearch.com/renewable-sector-published-reports/assessing-indias-pli-scheme-for-advanced-chemistry-cells-2
[44] https://documents1.worldbank.org/curated/en/772841622724981757/pdf/Electric-Mobility-in-India-Accelerating-Implementation.pdf

Sr. no.	Critical Mineral	Percentage	Major Import Sources (2020)
1	Lithium	100%	Chile, Russia, China, Ireland, Belgium
2	Cobalt	100%	China, Belgium, Netherlands, US, Japan
3	Nickel	100%	Sweden, China, Indonesia, Japan, Philippines
4	Vanadium	100%	Kuwait, Germany, South Africa, Brazil, Thailand
5	Niobium	100%	Brazil, Australia, Canada, South Africa, Indonesia
6	Germanium	100%	China, South Africa, Australia, France, US
7	Rhenium	100%	Russia, UK, Netherlands, South Africa, China
8	Beryllium	100%	Russia, UK, Netherlands, South Africa, China
9	Tantalum	100%	Australia, Indonesia, South Africa, Malaysia, US
10	Strontium	100%	China, US, Russia, Estonia, Slovenia
11	Zirconium(zircon)	80%	Australia, Indonesia, South Africa, Malaysia, US
12	Graphite(natural)	60%	China, Madagascar, Mozambique, Vietnam, Tanzania
13	Manganese	50%	South Africa, Gabon, Australia, Brazil, China
14	Chromium	2.5%	South Africa, Mozambique, Oman, Switzerland, Turkey
15	Silicon	<1%	China, Malaysia, Norway, Bhutan, Netherlands

Table X: The net import reliance for critical minerals of India (2020)

(Source: A report on 'Unlocking Australia-India Critical Minerals Partnership Potential' by Australian Trade and Investment Commission, July 2021)

India's battery and clean energy sectors remain highly dependent on imports for several critical minerals essential for lithium-ion battery manufacturing. As of 2020, the country relied entirely (100% import dependence) on imports of **lithium, cobalt, nickel, vanadium, niobium, germanium, rhenium, beryllium, tantalum, and strontium**. Import dependence was also significant for **zirconium (80%)**, **natural graphite (60%)**, and **manganese (50%)**, while **chromium (2.5%)** and **silicon (<1%)** were largely sourced domestically. These minerals are primarily imported from countries such as **China, Australia, Chile, Russia, South Africa, Indonesia, Brazil, and the United States**, highlighting India's exposure to global supply chain disruptions and geopolitical risks. The heavy reliance on imported critical minerals underscores the importance of developing domestic resource security, strengthening battery recycling infrastructure to recover valuable materials, and diversifying import sources to support the country's rapidly expanding electric mobility and energy storage sectors.

Mineral	**Domestic Availability**	**Status**
Lithium	5.9 million tonnes inferred resources (Jammu & Kashmir)	Exploration stage
Cobalt	Small associated deposits	Import-dependent
Nickel	Limited, low-grade deposits	Import-dependent

Graphite	Domestic occurrence; limited battery-grade processing	High import dependence
Manganese	~503.6 million tonnes of resources	Commercial production
Copper	Moderate reserves	Commercial production
Aluminium	Large bauxite reserves	Commercial production
Iron	Large reserves	Commercial production

Table X: Domestic Availability and Status of Critical Minerals for EV Batteries in India[45]

Government Initiatives: Recognising that access to critical minerals is fundamental to the country's clean energy transition and battery manufacturing ambitions, the Government of India has introduced several policy and institutional initiatives to reduce import dependence, strengthen domestic resource availability and diversify international supply chains. These initiatives collectively address the entire critical mineral value chain—from exploration and mining to processing, recycling and overseas asset acquisition.

1. National Critical Mineral Mission (NCMM) – 2025

The National Critical Mineral Mission (NCMM) is India's flagship programme for building a resilient critical mineral ecosystem. Approved by the Union Cabinet in January 2025, the mission has a financial outlay of ₹16,300 crore, with an additional ₹18,000 crore of expected investments from public sector undertakings and other stakeholders over seven years. Unlike previous initiatives that focused primarily on exploration, the NCMM adopts a value-chain approach by covering mineral exploration, mining, beneficiation, processing, recycling and recovery from end-of-life products.

Key targets include:

- 1,200 domestic exploration projects between 2024–25 and 2030–31.
- Acquisition of 26 overseas critical mineral assets through public sector enterprises.
- Development of critical mineral processing parks.
- Financial support for recycling and recovery of strategic minerals.

[45] https://ibm.gov.in/writereaddata/files/17501634866851601e3668eIMIG_2223_Annex_IIMYB.pdf, https://mines.gov.in/admin/download/6433da09a9f741681119753.pdf, https://www.pib.gov.in/PressReleasePage.aspx?PRID=1897799, https://www.niti.gov.in/sites/default/files/2023-07/Mine-to-market_NITI-Aayog-Publication_June-2023.pdf, https://www.iea.org/reports/the-role-of-critical-minerals-in-clean-energy-transitions

- Establishment of a Centre of Excellence on Critical Minerals to promote research and technology development.[46]

2. Identification of Critical Minerals

To prioritise strategic resource planning, the Ministry of Mines constituted an expert committee that identified 30 critical minerals essential for India's economic growth, energy transition and national security. Subsequently, 24 minerals were included in Part D of the First Schedule of the Mines and Minerals (Development and Regulation) Act, allowing the Central Government to directly auction mining leases for these strategically important resources. This policy aims to accelerate domestic exploration and improve regulatory coordination.

Examples include:

- Lithium
- Cobalt
- Nickel
- Graphite
- Rare Earth Elements
- Copper
- Molybdenum
- Vanadium[47]

3. Khanij Bidesh India Limited (KABIL)

Recognising that domestic mineral resources alone cannot satisfy future demand, the Government established Khanij Bidesh India Limited (KABIL) to secure overseas supplies of lithium, cobalt and other strategic minerals.

Major developments include:

- Agreement with CAMYEN (Argentina) for exploration and development of five lithium brine blocks covering approximately 15,703 hectares.
- Memorandum of Understanding with Australia's Critical Minerals Facilitation Office for joint due diligence and investment opportunities.
- Exploration and partnership discussions in Australia, Chile and several African countries to diversify India's mineral supply sources.

This initiative reduces India's dependence on a single geographic source while supporting long-term battery manufacturing.[48]

4. MMDR Amendment Act, 2023

The Mines and Minerals (Development and Regulation) Amendment Act, 2023 introduced significant reforms for deep-seated and critical minerals by enabling exploration licences and streamlining the auction process. These reforms are intended to encourage greater private-sector

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[46] https://www.pmindia.gov.in/en/news_updates/cabinet-approves-national-critical-mineral-mission-to-build-a-resilient-value-chain-for-critical-mineral-resources-vital-to-green-technologies-with-an-outlay-of-rs-34300-crore-over-seven-years/

[47] https://www.pib.gov.in/PressReleasePage.aspx?PRID=2120525

participation, accelerate exploration activities and improve the commercial viability of mining projects for strategic minerals such as lithium, cobalt, nickel and graphite.[48]

5. Geological Survey of India (GSI) Exploration Programme

The Geological Survey of India (GSI) has significantly expanded exploration activities for critical minerals under the National Critical Mineral Mission. According to the Ministry of Mines, GSI has been assigned 1,200 exploration projects between 2024–25 and 2030–31. Exploration activities have also identified lithium-bearing resources in Jammu & Kashmir and Rajasthan, strengthening India's long-term domestic resource potential.[48]

7.4 Evolution of EV in India and Battery Policies

EV Policy Timeline

1. **2013 — NEMMP 2020 Launched (ROADMAP)**
   - National Electric Mobility Mission Plan (NEMMP 2020) announced.
   - India's first long-term roadmap for electric mobility.
   - Set the foundation for demand incentives and manufacturing support.
2. **April 2015 — FAME I Begins (PHASE 1)**
   - ₹529 crore allocated.
   - Supported 2.7 lakh e-2W, e-3W and 465 e-buses.
   - Early charging stations in select cities.
   - Provided proof of concept: lower costs, better performance data.
3. **March 2019 — FAME I Concludes (PHASE 1 ENDS)**
   - Programme closed after four years.
   - Significant learnings for scaling EV adoption.
4. **April 2019 — FAME II Launched (PHASE 2)**
   - ₹10,000 crore — 11× larger than FAME I.
   - Incentives tied to Li-ion batteries.
   - Targets: 10 lakh e-2W, 5 lakh e-rickshaws, 55,000 four-wheelers, 7,000 e-buses, 2,700 charging stations.
   - Manufacturers faced localisation and subsidy delays.
5. **2021 — PLI Scheme: Autos & Components (MANUFACTURING)**
   - ₹26,058 crore approved.
   - Up to 18% incentives for EV & hydrogen fuel-cell models.
   - Triggered approx. ₹18,000 crore in new investments.
   - Suppliers scaled motor laminations & battery packs.
6. **September 2024 — PM E-Drive Approved (CURRENT SCHEME)**
   - ₹10,900 crore for 2024–2026.
   - Demand incentives: e-2W, e-3W, e-trucks, e-ambulances, e-truck, 14,028 e-buses.

- 22,100 fast chargers + 50,000+ charging points for scooters/rickshaws.
- Funding for modern testing & safety-certification facilities.

7.5 Recycling and Circular Economy Readiness

Battery recycling is emerging as a strategic pillar of India's battery ecosystem, supporting resource security, environmental sustainability and supply-chain resilience. Unlike conventional waste management, lithium-ion battery recycling enables the recovery of valuable critical minerals such as lithium, cobalt, nickel, manganese and copper, which can be reintroduced into battery manufacturing. As India accelerates electric vehicle adoption and battery energy storage deployment, the volume of end-of-life batteries is expected to increase significantly over the coming decades. Developing a robust recycling ecosystem can therefore reduce dependence on imported raw materials, lower greenhouse gas emissions associated with primary mining and strengthen domestic availability of battery-grade materials. According to RMI, a circular battery economy could create a ₹75,500 crore (approximately US$9 billion) market opportunity for India through battery reuse, repurposing and recycling while improving resource security.

Current capacity: India’s lithium-ion battery recycling sector remains nascent in terms of actual formal processing volumes, although announced capacity has expanded rapidly. A recent NITI-hosted assessment identifies more than 80,000 tonnes per year of announced recycling capacity, compared with approximately 15,000 tonnes of batteries expected to require formal recycling in 2025. The principal constraints are therefore collection, formal feedstock access, plant utilisation and the quality of recovered outputs, rather than nameplate capacity alone. Informal channels continue to handle a significant share of used batteries, often with limited traceability and inadequate safety controls.

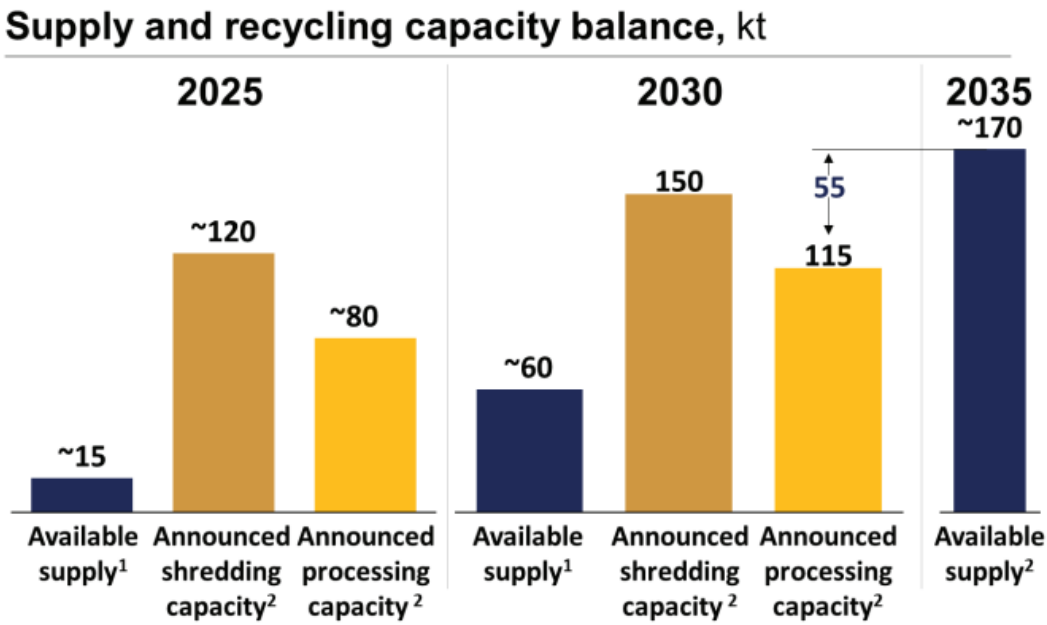


Figure X: Projected End-of-Life Lithium-ion Battery availability and recycling capacity in India

Regulatory framework: The Government of India has established a comprehensive regulatory framework to promote environmentally sound battery recycling through the Battery Waste Management Rules, 2022. These rules apply to all categories of batteries—including electric vehicle, automotive, industrial and portable batteries—and are based on the principle of Extended Producer Responsibility (EPR). Under this framework, producers are responsible for collecting waste batteries, ensuring their recycling or refurbishment through registered entities, and progressively incorporating recycled materials into new battery manufacturing. The Rules prohibit landfill disposal and incineration of waste batteries, establish a centralized online EPR portal for registration and compliance, and require producers to meet annual collection and recycling targets. These measures are intended to formalize India's recycling industry while improving traceability and resource recovery.[48]

Future scrap volumes: It is estimated that cumulative advanced-chemistry battery recycling potential of approximately 128 GWh between 2022 and 2030, of which around 59 GWh may originate from EV batteries. A more recent assessment projects India's end-of-life lithium-ion battery availability to rise from approximately 19,000 tonnes in 2023 to 233,000 tonnes by 2035. While announced recycling capacity currently exceeds formal feedstock availability, substantially

[48] https://niti.gov.in/sites/default/files/2026-01/Advancing-Circular-Economy-of-Waste-Electronic-and-Electrical-Equipment-Ewaste-and-Lithium-Ion-Batteries-in-India.pdf

stronger collection, logistics and processing systems will be required as EV and energy-storage batteries begin retiring at scale.[49]

Technology readiness: India's formal lithium-ion recycling industry predominantly uses mechanical pretreatment followed by hydrometallurgical recovery, while some operators currently produce and sell black mass without undertaking full refining. Advanced direct recycling, which preserves and regenerates cathode material, remains at an early stage and has not yet been commercially deployed at scale in India. The main technology challenge is achieving consistent, high-purity battery-grade recovery at competitive cost. This is particularly important for LFP batteries, whose lack of high-value nickel and cobalt makes recycling economics less attractive than for NMC and LCO chemistries.[50]

8.Closing the Loop: Black Mass, Urban Mining and Second-Life Batteries

8.1 Black Mass Opportunity

India's recycling industry today is concentrated at the pretreatment stage of the value chain — discharging, dismantling and mechanically processing end-of-life batteries into "black mass," a metal-concentrated intermediate powder containing recoverable lithium, nickel, cobalt and manganese. Many Indian operators currently stop here, selling black mass onward rather than completing the more technically demanding hydrometallurgical refining step needed to recover battery-grade materials domestically. This matters directly for the mineral-security argument, recycling capacity that stops at black mass does not by itself reduce import dependence — it simply relocates the refining step, and the value it captures, to whichever country has hydrometallurgical capacity, typically China.

This risk has sharpened recently. China lifted its ban on importing high-grade black mass in August 2025 and cut import tariffs on it from early 2026 — a policy shift explicitly aimed at pulling recyclable feedstock generated outside China toward Chinese refining facilities. Without domestic hydrometallurgical investment, India's growing black-mass supply is at risk of following the same trajectory as its raw mineral exports historically have: extracted or recovered domestically, refined elsewhere. Conversely, this represents a concrete, near-term investment opportunity — unlike upstream mining, hydrometallurgical refining capacity for already-recovered black mass could be

[49] https://www.niti.gov.in/sites/default/files/2022-07/ACC-battery-reuse-and-recycling-market-in-India_Niti-Aayog_UK.pdf , https://niti.gov.in/sites/default/files/2026-01/Advancing-Circular-Economy-of-Waste-Electronic-and-Electrical-Equipment-Ewaste-and-Lithium-Ion-Batteries-in-India.pdf

[50] https://www.niti.gov.in/sites/default/files/2023-07/Battery-book-july-web-version.pdf

built and scaled considerably faster, making it one of the more actionable near-term levers available to Indian policymakers and investors.[51]

8.2 Urban Mining

Framing battery recycling as urban mining — domestic resource creation rather than waste management — reframes the timing problem that dominates most recycling discussions. Two feedstocks are available well before India's EV fleet reaches significant end-of-life volumes:

- **Manufacturing scrap.** As gigafactory investment ramps up (Tata/Agratas, Amara Raja, Waaree, Adani), production scrap can run as high as 30-40% of total output during early ramp-up phases. This is a substantial, immediately available feedstock that does not depend on vehicles ageing out of service, and positions in-plant or co-located recycling as a natural complement to India's emerging manufacturing base.
- **Recoverable mineral volumes.** India's existing recycling-potential estimate can be translated into concrete mineral terms: recovering the batteries reaching end-of-life by 2030 could yield approximately 17.5 kilotonnes of lithium, 18.3 kilotonnes of nickel, and 15.9 kilotonnes of cobalt — figures that connect the abstract GWh recycling-capacity numbers directly to the import-substitution figures.

The National Critical Minerals Mission's ₹1,500 crore allocation specifically supporting mineral extraction from recycling is the clearest policy signal that urban mining is being treated as a distinct pillar of India's mineral strategy, not merely an environmental afterthought to primary mining.[52]

8.3 Second-Life Batteries

India's second-life battery ecosystem — repurposing EV batteries that have degraded 20-30% (and are therefore no longer suitable for vehicle use, but retain 70-80% of their original capacity) into stationary energy-storage applications — is moving from pilot to early-commercial stage faster than is often assumed, extending total battery lifetime from a typical 4-8 years in vehicle use to as much as 10-15 years overall.

Concrete examples now operating in India:

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[51] https://www.energy-storage.news/securing-indias-battery-supply-chain-is-more-critical-than-ever/

[52] https://www.energy-storage.news/securing-indias-battery-supply-chain-is-more-critical-than-ever

- **JSW MG Motor's "Project REVIVE"** has run multiple pilots repurposing MG ZS EV batteries into Battery Energy Storage Systems (BESS), partnering with Lohum, BatX, and LICO Materials; one iteration produced India's first high-voltage second-life BESS, built with an indigenously developed battery management system in partnership with Vision Mechatronics.
- **Lohum**, India's largest battery-material recycling and repurposing company, partners with both MG Motor India (targeting over 1 GW of batteries repurposed over three years from MG's EV fleet) and international OEMs including Audi and Mercedes-Benz for second-life applications.
- **Nunam**, a Berlin/Bangalore-based start-up backed by the Audi Environmental Foundation, has deployed second-life electric rickshaws in India powered by retired Audi e-tron test-fleet batteries — one of the first second-life pilots to put repurposed EV batteries on Indian roads.
- **Zipbolt**, working with Mercedes-Benz, has diagnosed and deployed over 50 repurposed battery systems for battery-swapping applications in e-rickshaws and electric scooters — directly relevant to Section 6's battery-swapping discussion.

The pattern across these examples is notable: India's second-life activity is currently driven primarily by global OEM partnerships (Audi, Mercedes-Benz, MG) working with a small number of specialised domestic firms, rather than by domestic automakers' own end-of-life fleets — a reflection of India's EV fleet still being too young, in aggregate, to generate large domestic second-life volumes independent of these partnerships. This is consistent with the broader global pattern discussed in Section 6.3, where second-life economics remain constrained by warranty/liability questions and by falling new-battery prices eroding the reuse business case — though India's stationary-storage demand growth provides a plausibly larger domestic offtake market for second-life batteries than many other countries currently have.[53]

9. Benchmarking Indian States : Leading States and Policy Models

9.1 The State Policy Landscape

State governments have become central actors in India's e-mobility transition, governing the transport, power distribution, manufacturing, and urban planning functions that shape EV adoption on the ground. As of June 2025, **29 states and 7 union territories had notified EV policies**, with four more under draft consideration — a sharp acceleration from just a handful of states five years

[53] https://wri-india.org/perspectives/second-charge-unlocking-second-life-potential-ev-batteries

earlier. This creates genuine sub-national competition and policy diversity, which NITI Aayog and WRI India now formally track through the India Electric Mobility Index (IEMI), launched in 2025 specifically to enable states to benchmark progress, identify gaps, and learn from each other's approaches.

9.2 The India Electric Mobility Index – Framework and National Rankings

The IEMI evaluates all 37 states and UTs across three weighted themes: Transport Electrification Progress (50% weight — private and commercial EV adoption, governance initiatives, purchase incentives), Charging Infrastructure Readiness (30% — charger density, capital subsidies, renewable energy integration), and EV Research and Innovation Status (20% — EV startups, patents, R&D initiatives), combining into a composite score out of 100. States are categorised as Frontrunners (score 65-99), Performers (50-64), or Aspirants (0-49).

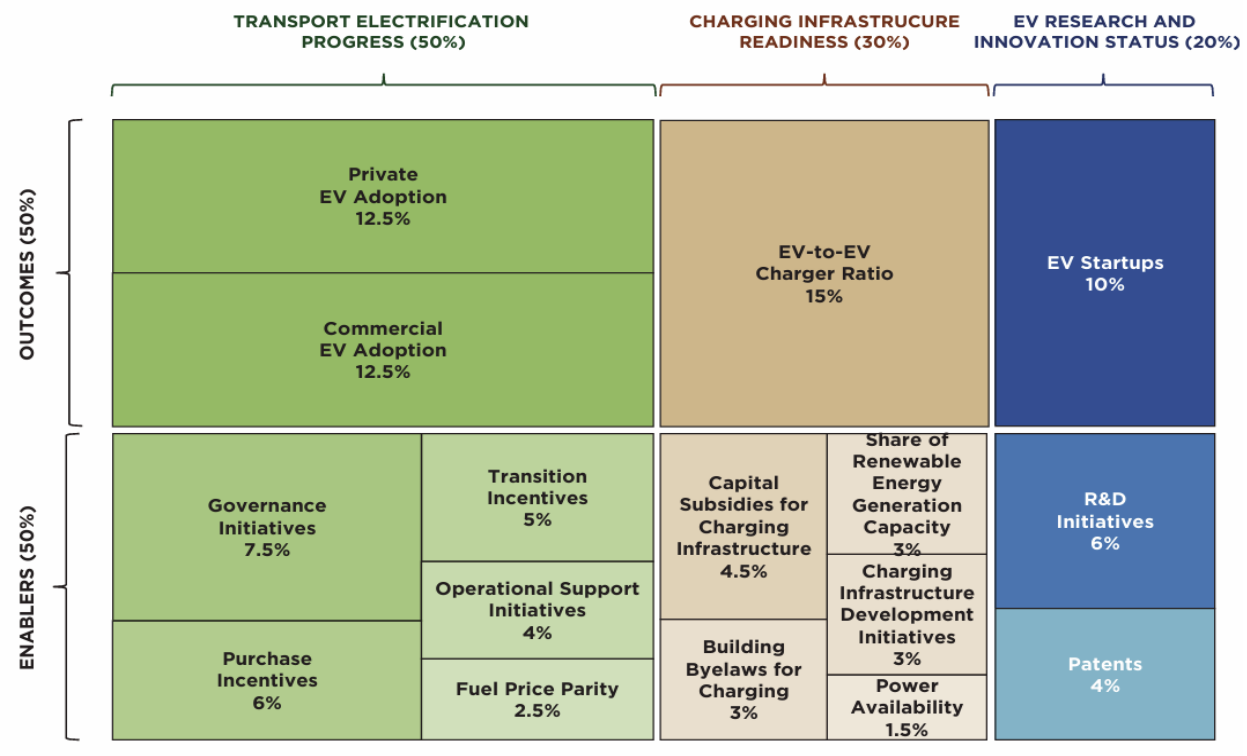


Figure : IEMI Theme and Indicator Weightages

Only six states/UTs — out of 36 — score as Frontrunners or Performers, underscoring how early-stage most of India's state-level EV ecosystem still is, even though nearly all states now have a notified policy on paper.

The IEMI score is determined by analysing the performance of each state/UT across all indicators, alongside the weight attributed to the indicators and the overarching themes of the index. This score reflects a state’s progress in advancing its e-mobility ecosystem. A higher IEMI score

indicates relatively strong performances across the three themes of EV adoption, charging infrastructure readiness and EV technology and innovation. Based on IEMI scores, states are categorised into frontrunners (leading with robust ecosystems), performers (making progress) and aspirants (requiring intervention).

Frontrunners: Delhi, Maharashtra and Chandigarh.

Performers: Karnataka, Tamil Nadu and Haryana

Aspirants: Odisha, Rajasthan, Chhattisgarh, Uttar Pradesh, Ladakh, Andhra Pradesh, Telangana, Punjab, Goa, Gujarat, Manipur, West Bengal, Kerala, Jharkhand, Tripura, Himachal Pradesh, Meghalaya, Mizoram, Madhya Pradesh, Bihar, Assam, Puducherry, Arunachal Pradesh, Lakshadweep, Uttarakhand, Andaman & Nicobar Islands, Sikkim, Nagaland, Jammu & Kashmir, and DD & DNH.[54]

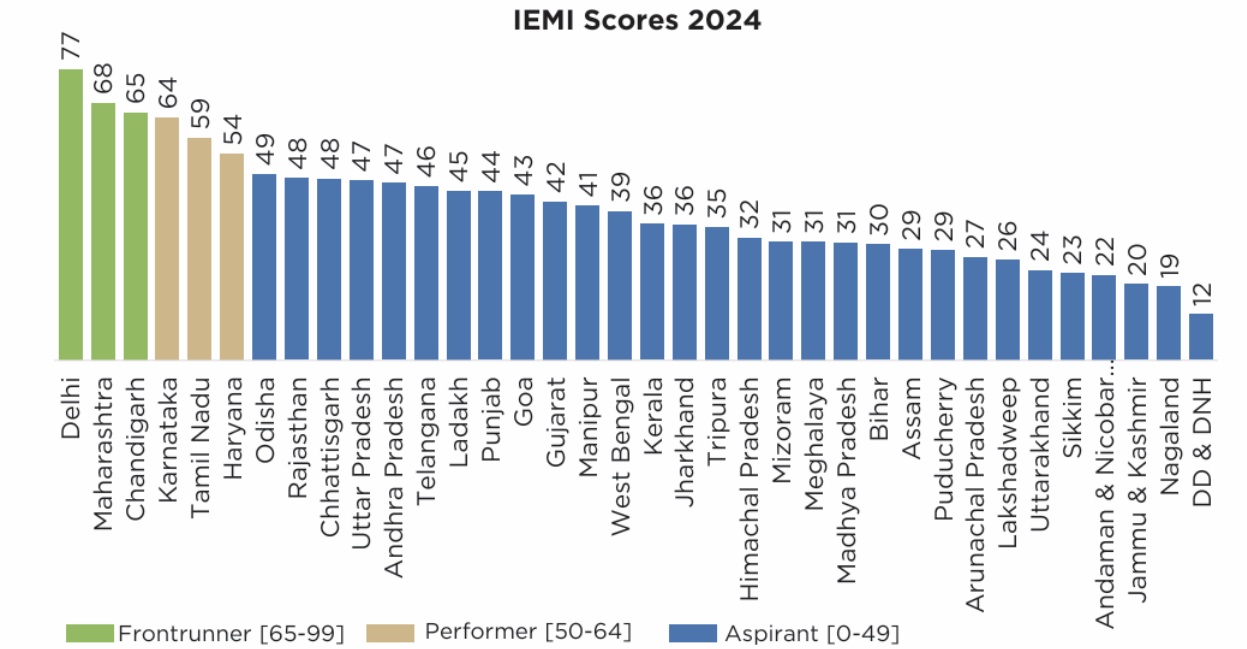


Figure : IEMI Score for 2024

9.3 Leading States and Policy Models

1.Delhi (Rank 1, score 77) leads on demand-side execution rather than manufacturing. It operates 2,500 electric buses — 33% of its public bus fleet, the highest share of any Indian state — and had disbursed ₹179 crore in EV purchase subsidies as of January 2024. Delhi was also the first state to extend subsidies to electric cycles, and topped the EV Research and Innovation theme nationally.

[54] India-Electric-Mobility-Index-2024-Report.pdf

Delhi's model demonstrates that consistent, well-funded demand-side incentives combined with strong governance (an EV-awareness portal, an interdepartmental EV committee) can outperform states with larger manufacturing investments on adoption outcomes.[54]

This model is now shifting. Delhi's original policy (2020) fell well short of its own targets — reaching only ~14% EV penetration against a 25% goal, and delivering just ~10% of its planned 45,000 public charging points before formally expiring in August 2024. In response, **Delhi EV Policy 2.0** — notified in July 2026 with an outlay of ₹15,000 crore, running through March 2030 — marks a deliberate shift from voluntary incentives to hard regulatory mandates: phased bans on new registrations of fossil-fuel three-wheelers and light delivery trucks from January 2027, and of fossil-fuel two-wheelers from April 2028, alongside a fleet-wide target of 30% electrification by March 2030.[55]

2. Maharashtra (Rank 2, score 68) combines strong Transport Electrification Progress with the country's most active startup ecosystem — 467 registered EV startups as of June 2025, the highest of any state.[54]This positions Maharashtra as India's leading hub for EV-adjacent innovation and entrepreneurship rather than large-scale cell manufacturing specifically.

Maharashtra's current EV Policy 2025-2030 — notified through a Transport Department resolution on 23 May 2025 — shows a clear shift from a purely adoption-focused approach toward one that also targets manufacturing and circularity:

- **Scale and targets**: backed by a ₹1,993 crore budget, the policy targets 30% of all new vehicle registrations being electric by 2030, with electric two-wheelers and passenger three-wheelers each expected to reach 40% of new registrations, and 50% of government and fleet-operator vehicles required to transition to electric.[56]
- **Purchase incentives**: ₹1,740 crore has been allocated to subsidise up to 1.75 lakh EV purchases across segments, with incentives ranging from 10-15% of a vehicle's base cost, and higher support (up to ₹20 lakh per unit) for transport vehicles and buses.[57]
- **Manufacturing push**: EV, battery, and component manufacturers qualify for "D+ category" industrial incentives — including stamp-duty and electricity-duty exemptions — explicitly extending the policy's ambition beyond adoption into local battery and component production, building on an earlier 2018/2021-era target of establishing 1 GW of in-state battery manufacturing capacity.[58]

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[55] https://www.impriindia.com/insights/policy-update/delhi-electric-vehicle-policy-2-0-2025/

[56] https://evreporter.com/maharashtra-electric-vehicle-policy-2025/
[57] https://www.mondaq.com/india/oil-gas-electricity/1635728/maharashtra-electric-vehicle-policy-2025
[58] https://www.deccanherald.com/india/maharashtra-unveils-new-ev-policy-sets-10-share-in-total-registrations-by-2025-1008353.html

- **Circular economy signal**: the policy introduces a digital battery passport to track manufacturing details, health status, usage history and end-of-life diagnostics for batteries sold in the state — a genuinely distinctive feature not present in any of the other states covered in this section, and directly relevant to your Section 8 discussion of black mass, urban mining and traceability.

3. Chandigarh (Rank 3, score 65): Chandigarh's Frontrunner ranking is unusual: as a small Union Territory rather than a state, its EV Policy 2022 aims explicitly to make Chandigarh a **"Model EV City,"** targeting one of the highest zero-emission-vehicle penetration rates of any Indian city within its five-year policy period (through September 2027).

Key design features:

- **Aggressive, revised targets**: the policy set an 18% overall EV-penetration target for FY2025-26, and had already exceeded its original electric-car target (achieving 20% registration share in 2023 against a 10% goal), though its two-wheeler targets were revised downward (from 70% to 35%, then further adjusted) after early implementation friction with dealers and residents.[59]
- **Mandatory government procurement**: from August 2023, all government departments and local bodies in Chandigarh are required to purchase only electric vehicles, with exceptions requiring prior approval — one of the more binding public-procurement mandates among Indian state/UT EV policies.[60]
- **Iterative, responsive policy design**: Chandigarh has amended its incentive structure multiple times in response to real-world uptake — increasing the two-wheeler subsidy from ₹5,000/kWh to ₹10,000/kWh (capped at ₹30,000) in 2025 after registrations fell once initial subsidy caps were reached, adding dedicated incentives for women riders, and expanding the number of electric four-wheelers eligible for incentives from 2,000 to 3,500 units. By August 2025, ₹35 crore in subsidies had been disbursed.[61]

4. Karnataka (Rank 4, score 64): Karnataka was the **first Indian state to launch a dedicated EV policy**, in 2017, strengthened in 2021 to improve investor economics, then substantially expanded through the **Clean Mobility Policy 2025-30**, notified 11 February 2025.

- **Scale of ambition**: the 2025-30 policy targets ₹50,000 crore in investment by 2030 and 100% electrification of public transport and government fleets by the same date —

[59] https://www.tribuneindia.com/news/chandigarh/big-respite-for-two-wheeler-buyers-522496/amp
[60] https://chandigarh.gov.in/sites/default/files/jan2022/crest20-evpolicy22-2009.pdf
[61] https://www.carnbikecafe.com/news/Electric-bike/chandigarh-expands-electric-vehicle-incentives-for-cars-two-wheelers

explicitly extending beyond battery electric vehicles to include hydrogen fuel-cell technology as a parallel clean-mobility pathway.[62]

- **Manufacturing incentives**: capital subsidies of up to 25% of total invested capex statewide (20% within Bengaluru Urban/Rural), covering the entire value chain — vehicle assembly, battery pack production, hydrogen fuel cells, and **battery recycling centres** explicitly named as an eligible incentive category, alongside 100% stamp-duty exemption and full reimbursement of land-conversion fees for agricultural-to-industrial conversion.[63]
- **Dedicated manufacturing clusters**: three "Clean Mobility Clusters" spanning 2,500+ acres are being developed in Gauribidanur, Dharwad, and Harohalli, offering ready-to-occupy land, built factories, incubation facilities, and testing labs — a deliberate attempt to co-locate OEMs, component suppliers, R&D centres, and testing infrastructure rather than leave manufacturers to source these independently.[64]
- **R&D support**: a 30% reimbursement of R&D costs (up to ₹1 crore) for enterprises meeting global benchmarks — one of the more direct state-level R&D subsidies among the states profiled in this section.[65]
- **Existing base**: Karnataka already has 2.5 lakh registered EVs and over 5,400 public charging stations (third-highest EV adoption in India), with a further 2,600 stations planned via public-private partnerships and fast-charging stations planned at 50km intervals on major highways (Bengaluru-Pune, Bengaluru-Mysuru).[66]

5. Tamil Nadu (Rank 5, score 59): Tamil Nadu is structurally different from every other state profiled here: it leads on manufacturing investment, not demand-side adoption or charging density.

- **Original 2019 ambition, revised in 2023:** Tamil Nadu was among the first states to release a dedicated EV policy (2019), targeting ₹50,000 crore in investment and 1.5 lakh jobs; the 2023 revision filled gaps in the original policy, particularly around upfront capital support and demand incentives for commercial vehicles.[67]
- **Battery-specific manufacturing incentives:** manufacturers setting up EV, battery, or component production qualify for a 20% capital subsidy on eligible fixed-asset investment (through end-2025), a 100% electricity-tax exemption on captive or TANGEDCO-purchased power for five years, 100% SGST reimbursement for 15 years (subject to

[62] https://bolt.earth/blog/karnataka-ev-policy-explained-what-it-means-for-ev-charging-infrastructure

[63] https://www.mercomindia.com/karnatakas-clean-mobility-policy-2025-30-targets-%E2%82%B9500-billion-investment

[64] https://investkarnataka.co.in/wp-content/uploads/2025/03/Government-of-Karnataka-Unveils-Clean-Mobility-Policy.pd

[65] https://www.mercomindia.com/karnatakas-clean-mobility-policy-2025-30-targets-%E2%82%B9500-billion-investmen

[66] https://ecozaar.in/blog/karnataka-electric-vehicle-policy

[67] https://ssinternational.in/tamil-nadu-ev-policy-2023

investment/employment criteria), and explicit advanced chemistry cell subsidies — one of the more comprehensive manufacturing-incentive stacks among the states covered.

- **Investment already realised:** the state attracted approximately ₹24,000 crore (~US$3.2 billion) in EV-sector investment interest over the five years preceding the 2023 policy, with a potential 48,000 jobs identified across sanctioned projects.[^n23] Ola Electric's ₹7,614 crore (~US$920 million) EV and 20 GWh battery-manufacturing hub near Hosur is the largest single commitment, alongside existing manufacturing presence from Ather Electric, Ampere, and TVS Motor, and battery/component makers BYD, Grinntech, Lucas-TVS, and Li Energy — together forming the Hosur-Krishnagiri-Dharmapuri corridor now emerging as a dedicated EV manufacturing cluster.[68]
- **Demand-side complement:** 100% road tax, registration-fee, and permit-fee exemptions for EVs (through 31 December 2025), and a target of 50% EV penetration across all vehicle categories by 2025 — an aggressive demand target intended to create a captive local market for the manufacturing base the policy is simultaneously building.[69]

6. Haryana (Rank 6, score 54): already covered in depth in your Section 9 — is notable in this national comparison for two specific distinctions that set it apart from the five states above:

- It was the top-ranked state nationally on Charging Infrastructure Readiness, despite a middling overall IEMI score (54, sixth among the profiled states).
- Its EV Policy 2022 is the **only policy among the six leading states with an explicit, binding battery-recycling requirement** — Large and Mega manufacturing units must establish battery disposal, recycling, and material-recovery facilities as a condition of receiving incentives, rather than recycling being addressed through a separate, voluntary scheme.[70]

7. Odisha (Rank 7, score 49): Odisha's original EV Policy 2021 aimed for 20% BEV share in all new registrations by 2025 — a target it fell well short of, reaching only 8.8% by mid-2025 despite offering purchase subsidies since 2021.

Rather than persist with an underperforming framework, Odisha issued a full policy overhaul, the **Odisha Electric Vehicle Policy 2025**, which:

- Sets a longer-horizon, more realistic target: **50% BEV penetration in new registrations by 2036**, rather than repeating an aggressive short-term target.[^n30]

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[68] https://coezet.iitm.ac.in/tamil-nadu-ev-policy-2023
[69] Tamilnadu.pdf
[70] https://investharyana.in/content/pdfs/EV%20Policy%202022.pdf

- Raises subsidy levels significantly — the previous 15% purchase subsidy structure is now supplemented with support reaching up to 30% in some segments, alongside new incentives specifically for battery-swapping infrastructure.
- Retains a manufacturing ambition on paper (an MoU-based approach to attracting lithium-ion battery manufacturers) but has not yet secured a comparable large-scale gigafactory commitment to the states.[71]

8. Rajasthan (Rank 8, score 48): Rajasthan's EV Policy 2022 (valid through 2027, largely unrevised as of late 2025) combines demand incentives with a specific manufacturing ambition: five dedicated **industrial corridors** for EV and battery manufacturing, offering a 25% capital subsidy on plant and machinery (capped at ₹50 lakh).[^n31]

- Concrete investment to date is component- and assembly-focused rather than cell manufacturing: Hero Electric committed ₹1,200 crore for a two-wheeler plant in Alwar district, and Saera Electric Auto opened an e-2W frame/component facility in Bhiwadi.[72]
- A ₹200 crore EV promotion fund, launched via a centralised digital subsidy portal in 2025, reimburses SGST and offers flat per-vehicle payments based on battery capacity — a somewhat unusual design (most states scale subsidy by vehicle price, not battery size specifically).
- **A distinctive, uncommon incentive**: alongside its capital subsidy, Rajasthan offers a 5% interest subsidy on term loans for plant and machinery, available for up to five years and capped at ₹1 crore per year — a financing-cost lever few other states in this comparison offer.
- **IISD's classification**: Rajasthan is grouped among India's "Emerging states" (alongside Uttar Pradesh and Madhya Pradesh) — states with comparatively lower targeted support for capital-intensive segments like cell manufacturing, but well-positioned for downstream assembly and component manufacturing given large available land parcels and dedicated EV parks. This is a more precise read than treating Rajasthan as simply "behind" the leading states — it has a defined, different role in the national picture.[73]

9. Chhattisgarh (Rank 9, score 48): Chhattisgarh's EV Policy 2022 targets a modest 15% BEV share of new registrations by 2027, but its manufacturing-incentive structure is unusually granular

[71] https://www.saurenergy.com/solar-energy-news/from-15-to-30-odisha-ev-policy-expands-subsidies-for-charging-and-battery-swaps-1047542
[72] https://jmkresearch.com/ev-manufacturing-in-rajasthan-gets-impetus-with-5-industrial-corridors-being-promoted-as-ev-battery-manufacturing-hubs
[73] https://www.iisd.org/system/files/2026-04/policies-electric-vehicles-battery-manufacturing-india.pdf

by enterprise size — capital subsidies scaled from 25% (Micro units, capped at ₹15 lakh) down to 10% (Large/Mega units, capped at ₹10-20 crore for the first two units in each category).

- The policy also offers 100% SGST reimbursement for EV manufacturing and a 25% capital subsidy for the state's first 300 fast-charging stations.[74]
- Unlike Tamil Nadu, Karnataka, or Andhra Pradesh, Chhattisgarh has not yet attracted a large, named gigafactory-scale battery investment — its manufacturing ambition remains structurally aimed at MSMEs and smaller-scale component makers rather than the large integrated cell plants defining the leading states in Section 9.3.
- **A real, named investment**: TWI Group has committed ₹1,650 crore (~US$195 million) to manufacture EV kits for two-wheelers in the state — Chhattisgarh's clearest concrete manufacturing investment to date, and a useful concrete example to cite alongside its capital-subsidy framework.[75]

10. Uttar Pradesh (Rank 10, score 47): Uttar Pradesh's EV Manufacturing and Mobility Policy 2022 is notable for the sheer size of its manufacturing incentives relative to most other Aspirant states:

- A **30% capital subsidy** (capped at ₹1,000 crore) is available for up to **two "Ultra Mega" battery projects**, each requiring a minimum investment of ₹1,500 crore and at least 1 GWh of production capacity — among the largest single-project subsidy ceilings offered by any Indian state.[76]
- The policy explicitly frames UP's existing MSME base across auto and battery component manufacturing as a foundation for anchor investment, aiming to position the state as a "global hub" for EV and battery manufacturing on the strength of its scale (240 million population, ~8% of national GDP) rather than a specialised technology cluster.
- Demand-side incentives are similarly large in absolute terms (a ₹500 crore purchase-subsidy budget, up to ₹1 lakh per electric four-wheeler) but capped to a fixed number of early purchases, meaning the effective subsidy window is narrower than the headline figures suggest.[77]a
- **A concrete, operational example finally landed:** GoodEnough Energy commissioned a 7 GWh battery energy storage system (BESS) gigafactory in Noida in January 2026 — described as India's largest operational battery storage plant, built with an investment exceeding ₹450 crore, with plans to scale to 25 GWh within three years. This is a meaningful data point since it shows UP's scale-based strategy beginning to land actual

[74] https://evreporter.com/chattisgarh-ev-policy/
[75] https://www.iisd.org/system/files/2026-04/policies-electric-vehicles-battery-manufacturing-india.pdf
[76] https://www.mercomindia.com/uttar-pradesh-announces-incentives-ev-battery-manufacturing
[77] https://upevsubsidy.in/pdf/1_A_UP_EV_Policy_2022_English_Version.pdf

projects, even if not yet the "Ultra Mega" cell-manufacturing project the 2022 policy specifically targets.[73]

11. Andra Pradesh (Rank 11, score 47): Andhra Pradesh has moved further and faster than any other Aspirant-tier state profiled here. Its **Sustainable Electric Mobility Policy 4.0 (2024-2029)**, effective December 2024, targets carbon neutrality in transport by 2047 and offers MSME investment subsidies of up to 45%, alongside 100% stamp-duty and registration-fee exemptions. At the time of the policy's launch, the state had only 601 public charging stations statewide — roughly one per 205 km of road — underscoring how early-stage its infrastructure base still was.[78]

- What has changed since is manufacturing investment, not adoption infrastructure: in 2025-26, Andhra Pradesh secured **Waaree Energies' ₹8,175 crore integrated lithium-ion battery gigafactory** in Rambilli, Anakapalli district — at 16 GWh, covering the full battery value chain from cells to packs to grid-scale BESS, this is described as India's largest single lithium-ion battery manufacturing investment, expected to create around 3,000 direct jobs.
- **The single most distinctive feature among all 14 states in IISD's comparison**: Andhra Pradesh is the *only* state offering a structured top-up to central government incentives — an additional 10% of the central PLI incentive amount, capped at 5% of fixed capital investment, allowing state and central support to stack directly. IISD explicitly identifies this as the reason Andhra Pradesh is becoming increasingly attractive specifically for capital-intensive cell manufacturing, ahead of other Aspirational-tier states. This is a stronger, more precise explanation for why Waaree's gigafactory landed in Andhra Pradesh than incentive scale alone.[73]
- **A second, smaller gigafactory confirms the pattern**: ChemVolt Global has separately committed ₹2,500 crore (~US$296 million) for a 5 GWh lithium-ion cell gigafactory in the state, phased with 1 GWh by Q3 2027 and reaching 5 GWh by Q2 2029 — indicating Waaree's investment is not an isolated win but the start of a genuine cell-manufacturing cluster.

12. Telangana (Rank 12, score 46): Telangana's EV & Energy Storage Policy targets $4 billion in investment and 1.2 lakh jobs by 2030, combining demand incentives (100% road tax/registration exemption for the first 2 lakh e-two-wheelers and first 5,000 private e-four-wheelers) with a manufacturing-incentive stack aimed squarely at battery and energy-storage-system makers: a 20% capital-investment subsidy (capped at ₹30 crore), 100% net SGST reimbursement, a 25% power-tariff discount, and 100% electricity-duty exemption, all time-capped for five years.[79]

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[78] https://cleanmobilityshift.com/policy-regulation/road-to-electric-decoding-andhra-pradeshs-strategy-to-boost-ev-adoption/

[79] https://keepingupwithev.substack.com/p/telangana-ev-and-energy-storage-policy

- Telangana's most significant asset, however, is a specific anchor investment already operating rather than newly announced: it hosts **Amara Raja's gigafactory in Mahbubnagar**, backed by a ₹9,500 crore investment commitment over a decade, with battery pack assembly already operational and the first 1 GWh phase of cell production underway, scalable toward 16 GWh by 2030 — positioning the state as India's established centre for large-format battery manufacturing.[80]
- **Telangana Mobility Valley** — four mega clusters spanning 3,000 acres in and around Hyderabad, including a 1,200-acre engineering and innovation cluster in Vikarabad district, two dedicated EV manufacturing clusters at Zaheerabad and Seetharampur, and a dedicated energy-storage cluster at Divitipally.[^n30a] IISD explicitly compares this hub-and-spoke model to Shenzhen's industrial-park approach in China, suggesting Telangana is deliberately importing lessons from more mature manufacturing ecosystems rather than replicating the single-large-subsidy approach most other states rely on.
- **T-Works and T-Fund** — T-Works is described as India's largest prototyping centre, covering EV components, vehicle assemblies, and battery prototyping; T-Fund is a state-backed co-investment vehicle that has already funded EV and battery-technology startups alongside private venture capital.[^n30a] This gives Telangana institutional R&D infrastructure — prototyping capacity and early-stage capital — that most other states profiled in this section lack entirely, and helps explain why an anchor investment like Amara Raja's did not remain an isolated project.

9.4 Cross-state Policy Comparison and Key Success Factors

Although state governments share the common objective of accelerating electric mobility, their policy priorities differ significantly depending on existing industrial capabilities, urban transport needs, fiscal capacity and economic structure. The leading states identified in the India Electric Mobility Index demonstrate that there is no single model for building a successful electric mobility ecosystem. Instead, different states have achieved progress by focusing on distinct strengths, including demand-side incentives, manufacturing investment, charging infrastructure, innovation ecosystems and regulatory interventions. These experiences provide valuable lessons for other states seeking to strengthen their own battery and EV ecosystems.[81]

[80] https://www.nsws.gov.in/s3fs/2021-08/Telangana%20EV%20policy.pdf

[81] India-Electric-Mobility-Index-2024-Report.pdf

State	Primary Strength	Distinctive Policy Feature	Key Lesson
Delhi	EV Adoption	Strong purchase incentives and public transport electrification	Demand-side incentives accelerate EV adoption.
Maharashtra	Innovation and circular economy	Digital Battery Passport and strong startup ecosystem	Manufacturing should be integrated with traceability, innovation and circular economy practices.
Chandigarh	Public procurement	Mandatory government EV procurement	Government leadership can stimulate early market demand.
Karnataka	Manufacturing ecosystem	Clean Mobility Clusters and R&D incentives	Industrial clusters supported by R&D incentives attract long-term investment.
Tamil Nadu	Manufacturing	Capital subsidies and integrated EV supply chain	Manufacturing competitiveness requires comprehensive industrial support and supply-chain integration.
Haryana	Charging infrastructure	Mandatory battery recycling requirement	Charging infrastructure and sustainability policies should evolve together.

10. Conclusion and Way Forward

India's battery sector should no longer be viewed solely through the lens of electric mobility. As this report has shown across the global value chain, chemistry choices, the global market landscape, recycling economics, and the state-level policies, batteries have become a foundational

industrial technology underpinning electric vehicles, renewable energy integration, grid-scale storage, telecommunications, consumer electronics, and data-centre infrastructure. The countries that currently dominate global battery manufacturing did not get there by building gigafactories alone; they built integrated ecosystems spanning critical minerals, chemical refining, active materials, cell manufacturing, battery management systems, testing infrastructure, and recycling. The lesson for India is the same one this report has returned to repeatedly: competitive advantage is created across the entire value chain, not at a single stage of it.

India enters this transition with real structural advantages — one of the world's fastest-growing battery markets, a large existing automotive industry, and an increasingly mature electronics manufacturing base. But demand alone does not guarantee industrial leadership. If India continues to rely primarily on imported cells, materials, and manufacturing equipment while limiting domestic activity to pack assembly and vehicle manufacturing, most of the economic value created by the battery transition will continue to accrue outside the country. The question this report set out to answer, in the end, is not whether India will become a major battery *market* — that outcome already looks assured — but whether it will also become a globally competitive battery manufacturing economy.

The central conclusion of this report is that India's battery strategy must shift from a capacity-expansion approach to an ecosystem-development approach. Policy success should not be measured by announced gigawatt-hours or the number of gigafactories under construction. India's cumulative cost disadvantage against China in EV and battery component manufacturing is estimated at nearly 10%, and net localization for high-cost EV components — batteries, motors, DC-DC converters, on-board chargers — remains below 20% nationally. The ACC PLI scheme, India's flagship instrument for closing this gap, had commissioned only 1.4 GWh of its 50 GWh target as of late 2025 — a 2.8% achievement rate, with zero incentives disbursed to any beneficiary.[40] This is not simply slow execution; it reflects a design mismatch. Output-linked incentives like the ACC PLI and PLI Auto reward realized production after capital has already been committed, but do little to de-risk the upfront capital intensity and long payback periods that define first-of-their-kind battery, cell, and cathode-active-material investments specifically — and PLI Auto's own record makes the same point differently: by early 2026, only 9% of its ₹25,938 crore outlay had actually reached beneficiaries, years after launch.[82]

The clearest, most actionable fix identified across the sources reviewed for this report is to supplement subsidy-based incentives with direct or co-investment capital — fund-of-funds structures, state-backed equity stakes, or blended public-private vehicles — for the midstream and upstream segments that conventional venture capital structurally avoids, since the 5 to10 year runway to commercial viability in battery "deep tech" exceeds typical private investment horizons. China's own battery-manufacturing rise was built substantially on this model — municipal and provincial government funds took direct equity stakes in early-stage EV and battery manufacturers,

[82] States in the Driver's Seat: Policies localizing electric vehicle and battery manufacturing in India

a mechanism Indian states have used only in isolated, small-scale form so far. Scaling this kind of patient, risk-absorbing capital, rather than adding further rounds of capital subsidy alone, is the single highest-leverage manufacturing-policy shift available to India in the next several years.

An equally important lesson from recent state-level policy experience is that India should avoid pursuing a uniform manufacturing strategy across all regions. Section 9 of this report showed that of 14 major automotive states, nearly all now offer a broadly similar stack of capital subsidies, land concessions, and tax exemptions — but only a handful have differentiated their incentive design around the value-chain segment their existing industrial base is actually suited to. States with deep, established automotive ecosystems — Tamil Nadu, Maharashtra, and Gujarat — are best positioned for integrated, full-value-chain localisation: Tamil Nadu's EV Special Manufacturing Package lets firms choose from a flexible incentive menu including a dedicated Advanced Chemistry Cell capital subsidy; Karnataka funds dedicated Clean Mobility Clusters offering shared testing and factory infrastructure; and Maharashtra channels support through its Thrust Sector Policy 2024, which explicitly designates lithium-ion, lithium titanium oxide, and hydrogen fuel-cell batteries as priority sectors. Gujarat's own chemical-industry and port infrastructure, distinct from these three, points it toward battery chemicals and export-oriented materials processing rather than assembly.

Aspirational states with strong incentives but thinner manufacturing bases are, sensibly, specialising rather than competing head-on with this leading cluster. Andhra Pradesh remains the only state offering a structured top-up to central PLI incentives, and it is not a coincidence that it has also landed the country's largest single battery-manufacturing commitment— policy complementarity, not incentive size alone, appears to be the deciding factor for capital-intensive investment location.[40] Telangana's parallel strength in electronics and semiconductors, backed by institutional infrastructure like T-Works and T-Fund, positions it well for battery management systems and research-intensive manufacturing rather than bulk cell production alone. On the evidence of Section 10, Haryana and Punjab sit in this same aspirational tier, not among the leading automotive states — Haryana's EV Policy 2022 offers a distinctive Seed and Conversion Fund to help existing MSME auto-component units pivot into EV and battery manufacturing, positioning it as a manufacturing- and NCR-demand-facing hub, while Punjab's policy builds in case-by-case flexibility to attract giga-scale investment and leans toward recycling, repurposing, and circular material recovery rather than cell manufacturing at scale. Neither state should attempt to replicate Tamil Nadu or Karnataka's integrated scale; both are better served deepening these already-distinctive, narrower roles.[82]

Emerging, land-rich states are more realistically positioned in downstream assembly than in first-of-their-kind cell manufacturing, at least in the near term. Uttar Pradesh's EV Manufacturing and Mobility Policy 2022 offers India's largest single-project subsidy ceiling but disburses it over 20 years after commercial operation begins — a structure better suited to attracting scale commitments than to solving early-stage capital risk; Rajasthan pairs a comparatively modest capital-subsidy ceiling with an uncommon 5% interest subsidy on term loans, a financing-cost

lever more suited to component and assembly investment than to capital-intensive cell manufacturing.

The remaining states in IISD's comparison illustrate the same logic from its weaker end, and are worth naming precisely because their scheme design reveals where India's state-level policy stack is still thin. Gujarat's own EV policy offers no direct manufacturing incentives, deferring instead to its general industrial policy; Madhya Pradesh's EV Policy 2025 similarly defers to broader industrial-promotion tools rather than battery-specific instruments, though its positioning around the National Automotive Test Tracks shows an emerging recognition that testing capacity can itself become a specialisation. Odisha's mineral and metallurgical base gives it a plausible long-term role in refining and precursor materials, but its current EV Policy 2021 covers only micro and small battery-manufacturing units — a structural mismatch, since cell manufacturers are almost never small enterprises — with the state instead pursuing battery-assembly investment through MoUs rather than a dedicated fiscal mechanism. West Bengal's EV Policy 2021 goes furthest in the same direction, not identifying manufacturing promotion as a policy objective at all. This spread — from Andhra Pradesh's deliberate incentive stacking at one end to West Bengal's near-absence of manufacturing policy at the other — is itself evidence that India's states are not yet operating as a coordinated national manufacturing strategy, but as 30-plus independent experiments of widely varying design quality. Rather than fourteen competing manufacturing ecosystems, India needs complementary regional clusters connected through integrated national supply chains, logistics networks, research partnerships, and common industrial standards — and consolidating the lessons from its strongest state-level experiments, rather than leaving each state to rediscover them independently, is arguably as important to India's manufacturing outlook as any single central scheme.[40]

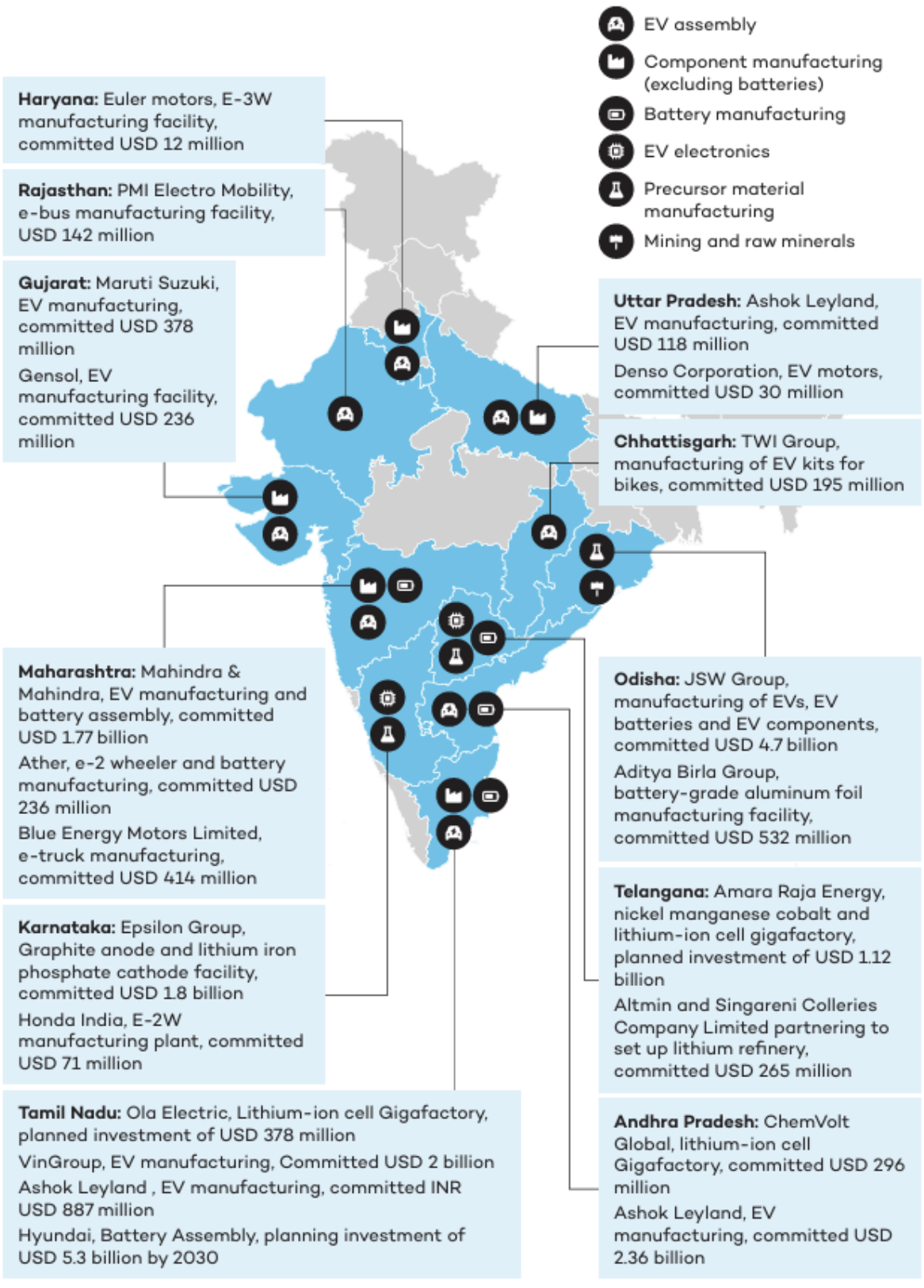


Figure : Recent EV and battery investment commitments across Indian states

Two further gaps deserve explicit attention because they are easy to overlook and expensive to leave unaddressed. Developing India's regional manufacturing clusters will require moving beyond fiscal incentives towards genuine ecosystem creation — plug-and-play industrial parks, common testing and certification laboratories, and stronger university-industry collaboration. Within this, India's innovation base remains particularly thin relative to its manufacturing ambition: of roughly 170,000 patents granted globally between 2010 and 2022 in electrical machinery, energy, and semiconductors, China accounted for 35% and India for barely 0.3%. A dedicated, IP-linked incentive — disbursed partly at filing and partly on commercial productization, modelled on the design-linked incentive scheme already used for semiconductors — would directly target this gap. Second, India's workforce-development pipeline has not kept pace with its manufacturing ambition: across the five Sector Skill Councils most relevant to EVs and batteries, only 21 total qualification packs exist for EV and battery work, just five specifically for batteries, and no dedicated Skill Council for batteries exists at all. Gigafactories cannot run on repair-and-maintenance training; they need deep-technical manufacturing skills India's vocational training system is not yet structured to produce at scale.

A resilient manufacturing ecosystem must also incorporate circular-economy principles from the outset, not as an afterthought. Recycling, refurbishment, repurposing, and material recovery should be understood as strategic industrial capabilities capable of strengthening resource security, not merely as environmental obligations. This report's findings on minerals and circularity point the same way as its findings on manufacturing: mineral ownership without refining access does not provide supply security (Section 7.3), and India's recycling capacity is currently ahead of its formal feedstock, with much of what is recovered stopping at black mass rather than completing domestic refining (Sections 6 and 8). India's Battery Waste Management Rules establish an important regulatory foundation, but a genuinely circular battery economy will require sustained investment in collection systems, digital battery traceability, black-mass processing, and second-life applications — gradually creating a secondary domestic source of lithium, nickel, cobalt, and graphite that reduces import dependence from a different direction than mining alone can.

India's battery opportunity ultimately extends far beyond manufacturing battery cells. It is an opportunity to build a globally competitive advanced-manufacturing platform integrating chemicals, metallurgy, electronics, precision engineering, and circular resource recovery under one industrial umbrella — and the country's long-term competitiveness will depend less on the number of factories it builds than on the strength of the ecosystem surrounding them. If India succeeds in converting its current demand advantage into manufacturing capability, deepens complementary state-level specialisation rather than duplicated ambition, strengthens upstream and midstream localisation through better-matched financing instruments, invests seriously in research and workforce skills, and embeds circularity into industrial design from the start, it has the potential to become not merely one of the world's largest battery markets, but one of its leading battery manufacturing economies. The defining measure of success should not be how many batteries India manufactures, but how much of that manufacturing genuinely belongs to

India — in value added, in technological capability, and in reduced exposure to the kind of external supply disruption this report opened by describing.